

Algorithm 3: Floating-Point Theorem Surfaces for SRHT and CountSketch Preconditioning

Scope

Algorithm 3 in the RandNLA survey applies random projection or preconditioning matrices to an input matrix A :

$$\Pi_L A, \quad A \Pi_R, \quad \Pi_L A \Pi_R.$$

The survey states this step for several random projection families [1]. This note records the final theorem surfaces for the closed signed-Hadamard/SRHT branch, the collision-free CountSketch actual-input branch, and the current implementation-facing CountSketch product-grid sparse-Gram branch. The SRHT part includes deterministic orthogonal-preconditioning identities, exact SRHT row-norm uniformization, uniform row sampling after preconditioning, and the floating-point transfer terms. The CountSketch part covers concrete collision-free and finite-cover Markov/Frobenius routes with sparse-apply and sparse-Gram floating-point arithmetic. These CountSketch routes are not the optimal input-sparsity subspace embedding theorem of Clarkson–Woodruff [4]. Gaussian and FJLT transforms are not included here.

Notation

Let m, n, s be positive integers, and let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns:

$$U^T U = I_n.$$

For a matrix $V \in \mathbb{R}^{m \times n}$, write V_{i*} for row i . If $V^T V = I_n$, define its leverage probability

$$\ell_i(V) = \frac{\|V_{i*}\|_2^2}{n}.$$

Let $H \in \mathbb{R}^{m \times m}$ be an exact orthogonal matrix whose entries are flat:

$$H^T H = I_m, \quad H_{ik}^2 = \frac{1}{m} \quad \text{for all } i, k.$$

Let $\omega = (\omega_1, \dots, \omega_m)$ be an exact Rademacher sign vector, meaning that the 2^m sign vectors in $\{\pm 1\}^m$ are equally likely. Set

$$D_\omega = \text{diag}(\omega_1, \dots, \omega_m), \quad V_\omega = H D_\omega U.$$

The random signs are the only randomness in V_ω .

After preconditioning, draw exact iid uniform row indices

$$J_1, \dots, J_s \in \{1, \dots, m\}, \quad \mathbb{P}[J_t = i] = \frac{1}{m}.$$

The exact uniform-row sketch of V_ω is

$$\tilde{V}_{\omega,J,tj} = \frac{(V_\omega)_{J_t j}}{\sqrt{s/m}},$$

and its exact sampled Gram matrix is

$$G_s(V_\omega) = \tilde{V}_{\omega,J}^T \tilde{V}_{\omega,J} = \frac{1}{s} \sum_{t=1}^s m (V_\omega)_{J_t*}^T (V_\omega)_{J_t*}.$$

The probability symbol $\mathbb{P}[\cdot]$ is always taken with respect to the exact product law of ω and $J = (J_1, \dots, J_s)$, unless a theorem explicitly restricts it to the sign law or the row-sampling law.

For symmetric matrices $X, Y \in \mathbb{R}^{n \times n}$, write $X \preceq Y$ when

$$x^T X x \leq x^T Y x \quad \text{for every } x \in \mathbb{R}^n.$$

Let $u \geq 0$ be the floating-point unit roundoff. When a length- r floating-point dot product or matrix product is charged, γ_r denotes the valid accumulated roundoff factor for that length, in the usual Higham model [7]. For a deterministic row-norm cap $R \geq 0$, define the uniform-row floating-point budgets

$$\begin{aligned} \tau_{\text{scale}}(R) &= n(2u + u^2)mR, \\ \tau_{\text{dot}}(R; s) &= n \gamma_s(1 + u)^2 mR, \end{aligned}$$

and

$$\tau_{\text{unif}}(R; s) = \tau_{\text{scale}}(R) + \tau_{\text{dot}}(R; s).$$

Here τ_{scale} charges rounded uniform row scaling and τ_{dot} charges the final rounded Gram dot products.

For $\theta > 0$, $L > 0$, and $\varepsilon > 0$, define

$$\beta_+(\theta, L) = \frac{\exp(\theta L) - \theta L - 1}{L^2}, \quad \beta_-(\theta) = \exp(\theta) - \theta - 1,$$

and the exact uniform-row tail budget

$$\begin{aligned} \Phi(\theta, L, \varepsilon, s, n) &:= e^{-\theta s \varepsilon} n e^{s \beta_+(\theta, L)(L^2+1)} \\ &\quad + e^{-\theta s \varepsilon} n e^{s \beta_-(\theta)(L^2+1)}. \end{aligned}$$

The parameter ε is the exact user-chosen subspace-embedding distortion tolerance. It is not computed by Algorithm 3 and therefore carries no floating-point error term. Once ε is chosen, the sample size s and Chernoff parameter θ must make the exact sampling budget $\Phi(\theta, L, \varepsilon, s, n)$ small enough. A standard constant-distortion instantiation is $\varepsilon = 1/2$; keeping ε symbolic gives the usual accuracy-versus-sample-size tradeoff. For a dimension parameter q , row-bound parameter L , distortion tolerance ε , and sampling failure budget δ , define the exact sample-count rule

$$s_{\min}(L, \varepsilon, \delta; q) = \min \{k \in \mathbb{N}_{>0} : \exists \theta > 0 \text{ such that } \Phi(\theta, L, \varepsilon, k, q) \leq \delta\}.$$

Thus s is an exact integer design parameter: the algorithm draws exactly s iid uniform row indices after this value has been chosen. The choice of s is not a floating-point computation in the theorem. The usual matrix-Chernoff scaling [6] gives the sufficient order

$$s_{\min}(L, \varepsilon, \delta; q) = O\left(\frac{L}{\varepsilon^2} \log \frac{q}{\delta}\right),$$

with absolute constants in the standard $0 < \varepsilon < 1$ regime. For the source-sharp SRHT row-norm radius, define

$$S(t) = \sqrt{\frac{n}{m}} + t, \quad R(t) = S(t)^2, \quad L(t) = mR(t), \quad \tau_{\text{SRHT}}(t) = \tau_{\text{unif}}(R(t); s).$$

What Is Exact and What Is Computed

- Exact arithmetic objects: the input matrix A as handed to Algorithm 3, any analysis-only orthonormal basis U for a column space, H , D_ω , V_ω , the Rademacher sign law, the uniform row law, the exact sketch $\tilde{V}_{\omega,J}$, and exact Gram matrices such as $G_s(V_\omega)$. These are the mathematical reference objects in the probability bounds.
- Exact integer and Boolean objects: a Sylvester/Walsh sign pattern generated from bit parities is an exact deterministic sign table. Its entries are ± 1 exactly. The scale factor $\sqrt{1/m}$, if computed in floating point, is not part of that exact integer/Boolean generation. The sample count s is an exact positive integer chosen from the sampling-budget rule $s_{\min}$; the theorem does not model floating-point error in choosing this integer.
- Floating-point objects after an exact V_ω is supplied: the uniform denominator $\sqrt{s/m}$ may be replaced by a computed $\hat{d}$, the sampled rows may be scaled by rounded division, and the final Gram entries may be computed by rounded dot products. These operations are charged by denominator and τ_{unif} budgets.
- Floating-point objects used to form V_ω : an implementation may store or generate $\hat{H}$, store realized signs $\hat{\omega}$, form $\hat{P}_\omega \approx H D_\omega$, use a computed input basis $\hat{U}$, and form $\hat{V}_\omega = \text{fl}(\hat{P}_\omega \hat{U})$. These quantities are computable in floating point only after explicit entrywise or normwise certificates are supplied.
- Floating-point objects used on the actual input matrix A : the materialized SRHT preconditioner $\hat{P}_\omega$, the product $\hat{Y}_\omega = \text{fl}(\hat{P}_\omega A)$, rounded uniform-row scaling, and the rounded sampled Gram matrix are computed. The exact comparison objects are $Y_\omega = P_\omega A$, the exact scaled sample from Y_ω , and $A^T A$.
- A QR, SVD, singular-vector, or basis routine that produces $\hat{U}$ is not part of the SRHT probability theorem. Its error enters through a separate matrix certificate for $\hat{U} - U$. Without that certificate, the theorem remains an exact- U statement.

Deterministic Orthogonal Preconditioning

Theorem 1 (Exact Frobenius stability). *Let $A \in \mathbb{R}^{m \times n}$. If $\Pi_L \in \mathbb{R}^{m \times m}$ and $\Pi_R \in \mathbb{R}^{n \times n}$ are exact orthogonal matrices, then*

$$\|\Pi_L A\|_F = \|A\|_F, \quad \|A \Pi_R\|_F = \|A\|_F, \quad \|\Pi_L A \Pi_R\|_F = \|A\|_F.$$

Theorem 2 (Exact orthonormal-column and leverage-denominator invariance). *Let $U \in \mathbb{R}^{m \times n}$ satisfy $U^T U = I_n$. If $\Pi_L \in \mathbb{R}^{m \times m}$ and $\Pi_R \in \mathbb{R}^{n \times n}$ are exact orthogonal matrices, then*

$$(\Pi_L U)^T (\Pi_L U) = I_n, \quad (U \Pi_R)^T (U \Pi_R) = I_n, \quad (\Pi_L U \Pi_R)^T (\Pi_L U \Pi_R) = I_n.$$

Consequently the leverage denominator remains exactly n :

$$\sum_{i=1}^m \|(\Pi_L U)_{i*}\|_2^2 = \sum_{i=1}^m \|(U \Pi_R)_{i*}\|_2^2 = \sum_{i=1}^m \|(\Pi_L U \Pi_R)_{i*}\|_2^2 = n.$$

Theorem 3 (Exact signed-Hadamard orthogonal preprocessing). *Let $H \in \mathbb{R}^{m \times m}$ be exact orthogonal, and let $\omega \in \{\pm 1\}^m$. Then D_ω and HD_ω are orthogonal. For every $U \in \mathbb{R}^{m \times n}$ with $U^T U = I_n$,*

$$(HD_\omega U)^T (HD_\omega U) = I_n, \quad \sum_{i=1}^m \|(HD_\omega U)_{i*}\|_2^2 = n.$$

Under the exact Rademacher law, this event has probability one.

Uniform Row Denominator

Theorem 4 (Computed uniform-row denominator certificate). *Let*

$$d = \sqrt{s/m}, \quad \hat{d} \neq 0, \quad d \neq 0, \quad |\hat{d} - d| \leq \eta_d.$$

For an exact row value V_{ij} , define the exact scaled value

$$B_{ij} = V_{ij}/d$$

and the floating-point scaled value

$$\hat{B}_{ij}^{\hat{d}} = \text{fl}(V_{ij}/\hat{d}).$$

Then

$$\left| \hat{B}_{ij}^{\hat{d}} - B_{ij} \right| \leq \left| V_{ij}/\hat{d} \right| u + \frac{|V_{ij}| \eta_d}{|\hat{d}| |d|}.$$

The row index is still drawn from the exact uniform law $1/m$. Only the non-probability scalar $\hat{d}$ and the final rounded division are charged.

Theorem 5 (Closed denominator routines). *Assume $0 < u < 1$, $m > 0$, $s > 0$, and $d = \sqrt{s/m}$. The preceding certificate applies with the following choices of η_d :*

<i>exact denominator</i>	$\eta_d = 0,$
$\hat{d} = \text{fl}(\sqrt{s/m})$ with exact input s/m	$\eta_d = d u,$
$\hat{x} = \text{fl}(s/m), \hat{d} = \text{fl}(\sqrt{\hat{x}})$	$\eta_d = d(\sqrt{1+u} u + u),$
$\hat{r} = \text{fl}(1/m), \hat{x} = \text{fl}(s\hat{r}), \hat{d} = \text{fl}(\sqrt{\hat{x}})$	$\eta_d = d \left(\sqrt{(1+u)(1+u)} u + 2u + u^2 \right),$
$\hat{s}_0 = \text{fl}(\sqrt{s}), \hat{m}_0 = \text{fl}(\sqrt{m}), \hat{d} = \text{fl}(\hat{s}_0/\hat{m}_0)$	$\eta_d = d \frac{3u + u^2}{1 - u},$
$\hat{s}_0 = \text{fl}(\sqrt{s}), \hat{m}_0 = \text{fl}(\sqrt{m}), \hat{r}_m = \text{fl}(1/\hat{m}_0), \hat{d} = \text{fl}(\hat{s}_0 \hat{r}_m)$	$\eta_d = d \frac{4u + 3u^2 + u^3}{1 - u}.$

Each line charges a different floating-point way to compute the same exact denominator $d = \sqrt{s/m}$. None of these routines changes the uniform row law. The final SRHT implementation-facing theorems below use the last routine,

$$\hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{m}))), \quad |\hat{d}_\star - d| \leq d \frac{4u + 3u^2 + u^3}{1 - u}.$$

The stored-sign CountSketch endpoint at the end of this note is also instantiated with the four rounded alternatives above: exact-input square root, divide-then-square-root, reciprocal-multiply-then-square-root, and square-root-divide-square-root. The exact-denominator line remains an analysis/infrastructure specialization rather than a floating-point denominator computation.

Exact SRHT Uniformization

Exact baseline (Source-sharp signed-Hadamard row norms). Assume $m > 0$, $U^T U = I_n$, and $H_{ik}^2 = 1/m$ for every i, k . For every $t > 0$,

$$\mathbb{P}_\omega \left[\forall i, \quad \|(HD_\omega U)_{i*}\|_2 \leq \sqrt{\frac{n}{m}} + t \right] \geq 1 - m \exp\left(-\frac{mt^2}{8}\right).$$

Equivalently,

$$\mathbb{P}_\omega \left[\forall i, \quad \ell_i(HD_\omega U) \leq \frac{(\sqrt{n/m} + t)^2}{n} \right] \geq 1 - m \exp\left(-\frac{mt^2}{8}\right).$$

This is the signed-Hadamard/SRHT row-norm bound corresponding to Tropp's SRHT analysis [2]; the CACM Algorithm 3 discussion uses this kind of random preprocessing to make subsequent uniform row sampling effective [1].

Theorem 6 (Logarithmic SRHT preprocessing radius). Let $0 < \delta_{\text{pre}} < m$, and set

$$t_\delta = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}.$$

Then

$$\mathbb{P}_\omega \left[\forall i, \quad \|(HD_\omega U)_{i*}\|_2^2 \leq \left(\sqrt{\frac{n}{m}} + t_\delta \right)^2 \right] \geq 1 - \delta_{\text{pre}}.$$

Exact baseline (Coordinate-Hoeffding fallback). Let $B \geq 0$, $\lambda > 0$, and define

$$\eta_{\text{coord}}(B, \lambda, m) = 2 \exp(-\lambda B) \exp\left(\frac{\lambda^2}{2m}\right).$$

If

$$\sum_{i=1}^m \sum_{j=1}^n \eta_{\text{coord}}(B, \lambda, m) \leq \delta_{\text{pre}},$$

then

$$\mathbb{P}_\omega \left[\forall i, \quad \|(HD_\omega U)_{i*}\|_2^2 \leq nB^2 \right] \geq 1 - \delta_{\text{pre}}.$$

When H is orthogonal and $n > 0$, the same event implies

$$\ell_i(HD_\omega U) \leq B^2 \quad \text{for every } i.$$

This fallback is weaker than the source-sharp SRHT radius, but it has the same exact/computed separation: the signs are random exact objects, while all later floating-point arithmetic is charged separately.

Exact Uniform Row Sampling After Preconditioning

Theorem 7 (Uniform one-step Loewner bound). *Let $V \in \mathbb{R}^{m \times n}$ have orthonormal columns, and define*

$$X_i(V) = m V_{i*}^T V_{i*} \in \mathbb{R}^{n \times n}.$$

Then $X_i(V) \succeq 0$ and

$$\frac{1}{m} \sum_{i=1}^m X_i(V) = I_n.$$

If $\|V_{i}\|_2^2 \leq R$, then*

$$X_i(V) \preceq mR I_n.$$

Exact baseline (Uniform row sample-average concentration). *Let $V \in \mathbb{R}^{m \times n}$ have orthonormal columns. Suppose that for every row i ,*

$$X_i(V) \preceq L I_n, \quad m \|V_{i*}\|_2^2 \leq L.$$

If

$$\Phi(\theta, L, \varepsilon, s, n) \leq \delta_{\text{sample}},$$

then

$$\mathbb{P}_J[G_s(V) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V) - I_n) \preceq \varepsilon I_n] \geq 1 - \delta_{\text{sample}}.$$

This is the matrix trace-MGF row-sampling bound used in the SRHT composition; the matrix concentration lineage is represented by Tropp's matrix tail bounds [6].

Theorem 8 (Exact source-sharp SRHT plus uniform row sampling). *Let $t > 0$. Assume*

$$m \exp\left(-\frac{mt^2}{8}\right) \leq \delta_{\text{pre}}$$

and

$$\Phi(\theta, L(t), \varepsilon, s, n) \leq \delta_{\text{sample}}.$$

Then, under the exact product law of Rademacher preprocessing and iid uniform row sampling,

$$\mathbb{P}[G_s(V_\omega) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V_\omega) - I_n) \preceq \varepsilon I_n] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

Theorem 9 (Exact logarithmic SRHT plus uniform row sampling). *Let $0 < \delta_{\text{pre}} < m$, and set*

$$t_\delta = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}.$$

If

$$\Phi(\theta, L(t_\delta), \varepsilon, s, n) \leq \delta_{\text{sample}},$$

then

$$\mathbb{P}[G_s(V_\omega) - I_n \preceq \varepsilon I_n \text{ and } -(G_s(V_\omega) - I_n) \preceq \varepsilon I_n] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

Floating-Point Uniform Row Sampling

In this section $V_\omega = HD_\omega U$ is still the exact mathematical preconditioned basis. Let

$$\hat{G}_{\omega,J}^{\text{exactV}}$$

denote the Gram matrix whose sampled rows are taken from this exact V_ω , whose row scaling is carried out by floating-point division, and whose entries are computed by floating-point dot products.

Theorem 10 (Fixed row-norm floating-point radius). *Assume γ_s is valid. If every sampled exact row satisfies*

$$\|(V_\omega)_{J_t^*}\|_2^2 \leq R \quad (t = 1, \dots, s),$$

then

$$\left\| \hat{G}_{\omega,J}^{\text{exactV}} - G_s(V_\omega) \right\|_F \leq \tau_{\text{unif}}(R; s).$$

Consequently, any exact two-sided Loewner event with radius ε transfers to the computed Gram matrix with radius $\varepsilon + \tau_{\text{unif}}(R; s)$.

Theorem 11 (Source-sharp SRHT floating-point uniform row sampling). *Let $t > 0$. Assume*

$$m \exp\left(-\frac{mt^2}{8}\right) \leq \delta_{\text{pre}}, \quad \Phi(\theta, L(t), \varepsilon, s, n) \leq \delta_{\text{sample}},$$

and assume γ_s is valid. Then

$$\mathbb{P}\left[-(\varepsilon + \tau_{\text{SRHT}}(t))I_n \preceq \hat{G}_{\omega,J}^{\text{exactV}} - I_n \preceq (\varepsilon + \tau_{\text{SRHT}}(t))I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

The logarithmic version substitutes

$$t = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}$$

whenever $0 < \delta_{\text{pre}} < m$.

Remark. The probability lower bound is unchanged from the exact SRHT theorem. The only change is the deterministic additive radius

$$\tau_{\text{SRHT}}(t) = n(2u + u^2)mR(t) + n\gamma_s(1 + u)^2mR(t).$$

The first term is row-scaling error and the second term is dot-product error.

Fully Computed Modeled SRHT Bounds

This section fixes a concrete computed path and gives actual error bounds for that path. The path is a materialized scaled-pattern SRHT preconditioner: the sign pattern is generated exactly by integer/Boolean arithmetic, the normalization scale is rounded, the realized Rademacher signs are stored by a rounded copy routine, the preconditioner matrix is formed by rounded products, and the preconditioned matrix or sampled Gram matrix is computed in floating point. The probability laws are still exact mathematical laws.

The closed fully computed path below does not claim to cover an arbitrary computed QR/SVD basis routine, an arbitrary FHT memory layout, or an arbitrary vectorized overwrite schedule. Those paths are not part of the final theorem unless their concrete roundoff recurrences are reduced to the same displayed computed quantities.

For any transform dimension N , define

$$\hat{p}_N = \frac{(1 + \gamma_N)(1 + u)^2}{\sqrt{N}},$$

$$\alpha_N = \frac{\gamma_N(1 + u)^2 + u(1 + u) + u}{\sqrt{N}},$$

and

$$\kappa_N = \gamma_N \hat{p}_N + \alpha_N.$$

Here $\hat{p}_N$ bounds the magnitude of every computed materialized SRHT entry, α_N bounds the absolute error of every computed materialized SRHT entry, and κ_N is the coefficient that appears when multiplying one such computed preconditioner by a matrix. These quantities are small when the usual accumulated factors γ_N and the unit roundoff u are small; the $1/\sqrt{N}$ scaling is the Hadamard normalization.

Theorem 12 (Computed materialized scaled-pattern SRHT preconditioner). *Let $N > 0$, and let $S \in \mathbb{R}^{N \times N}$ be an exact sign-pattern table:*

$$S_{ij}^2 = 1.$$

For the Sylvester/Walsh SRHT, S is the exact bit-parity sign table and is therefore generated with exact integer/Boolean arithmetic. Define the exact and computed scales

$$c_N = \sqrt{1/N}, \quad \hat{c}_N = \text{fl}(\sqrt{1/N}),$$

the exact and computed scaled tables

$$H_{ij} = c_N S_{ij}, \quad \hat{H}_{ij} = \hat{c}_N S_{ij},$$

and the scale-table error radius

$$E_{ij}^H = c_N u |S_{ij}|.$$

Then

$$|\hat{H}_{ij} - H_{ij}| \leq E_{ij}^H, \quad H_{ij}^2 = \frac{1}{N}.$$

Let $\sigma \in \{\pm 1\}^N$ be the exact realized Rademacher sign vector. The implementation stores it by one of the rounded copy routines

$$\hat{\sigma}_i = \text{fl}(\sigma_i \cdot 1), \quad \hat{\sigma}_i^+ = \text{fl}(\sigma_i + 0), \quad \hat{\sigma}_i^- = \text{fl}(\sigma_i - 0).$$

For any one of these three stored sign vectors, write the stored vector as $\hat{\sigma}^\rho$. Since $|\sigma_i| = 1$, its sign-storage error is

$$|\hat{\sigma}_i^\rho - \sigma_i| \leq E_i^{\sigma, \rho}, \quad E_i^{\sigma, \rho} = u.$$

Set

$$D = \text{diag}(\sigma), \quad \hat{D}^\rho = \text{diag}(\hat{\sigma}^\rho), \quad P = HD, \quad \hat{P}^\rho = \text{fl}(\hat{H}\hat{D}^\rho).$$

Under the floating-point model with valid γ_N , the computed materialized preconditioner satisfies

$$|\hat{P}_{ij}^\rho| \leq \hat{p}_N, \quad |\hat{P}_{ij}^\rho - P_{ij}| \leq \alpha_N.$$

Thus the bound charges the rounded scale, rounded sign storage, and rounded formation of the materialized HD preconditioner, but it exposes the result through the two interpretable scalars $\hat{p}_N$ and α_N .

Theorem 13 (Computed Algorithm 3 preconditioned matrices). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix for Algorithm 3. Let $S^L \in \mathbb{R}^{m \times m}$ and $S^R \in \mathbb{R}^{n \times n}$ be exact sign-pattern tables:*

$$(S_{ik}^L)^2 = 1, \quad (S_{kj}^R)^2 = 1.$$

Let $\sigma^L \in \{\pm 1\}^m$ and $\sigma^R \in \{\pm 1\}^n$ be the exact realized Rademacher sign vectors. Store each sign by one of the rounded copy routines $\text{fl}(\sigma_i \cdot 1)$, $\text{fl}(\sigma_i + 0)$, or $\text{fl}(\sigma_i - 0)$, and call the stored signs $\hat{\sigma}^L$ and $\hat{\sigma}^R$. Then

$$|\hat{\sigma}_i^L - \sigma_i^L| \leq u, \quad |\hat{\sigma}_j^R - \sigma_j^R| \leq u.$$

Define the exact and computed left scale tables

$$c_m = \sqrt{1/m}, \quad \hat{c}_m = \text{fl}(\sqrt{1/m}),$$

$$H_{ik}^L = c_m S_{ik}^L, \quad \hat{H}_{ik}^L = \hat{c}_m S_{ik}^L, \quad E_{ik}^{H,L} = c_m u |S_{ik}^L|.$$

Define the exact and computed right scale tables

$$c_n = \sqrt{1/n}, \quad \hat{c}_n = \text{fl}(\sqrt{1/n}),$$

$$H_{kj}^R = c_n S_{kj}^R, \quad \hat{H}_{kj}^R = \hat{c}_n S_{kj}^R, \quad E_{kj}^{H,R} = c_n u |S_{kj}^R|.$$

Set

$$P_L = H^L \text{diag}(\sigma^L), \quad \hat{P}_L = \text{fl}(\hat{H}^L \text{diag}(\hat{\sigma}^L)),$$

$$P_R = H^R \text{diag}(\sigma^R), \quad \hat{P}_R = \text{fl}(\hat{H}^R \text{diag}(\hat{\sigma}^R)).$$

The entrywise preconditioner radii are

$$|\hat{P}_{L,ik} - P_{L,ik}| \leq \alpha_m, \quad |\hat{P}_{R,kj} - P_{R,kj}| \leq \alpha_n,$$

and the computed preconditioner entries satisfy

$$|\hat{P}_{L,ik}| \leq \hat{p}_m, \quad |\hat{P}_{R,kj}| \leq \hat{p}_n.$$

For left preconditioning, define

$$B_L = P_L A, \quad \hat{B}_L = \text{fl}(\hat{P}_L A),$$

then

$$|(\hat{B}_L)_{ij} - (B_L)_{ij}| \leq \kappa_m \|A_{*j}\|_1 \leq \kappa_m \sqrt{m} \|A_{*j}\|_2.$$

For right preconditioning, define

$$B_R = A P_R, \quad \hat{B}_R = \text{fl}(A \hat{P}_R),$$

then

$$|(\hat{B}_R)_{ij} - (B_R)_{ij}| \leq \kappa_n \|A_{i*}\|_1 \leq \kappa_n \sqrt{n} \|A_{i*}\|_2.$$

For two-sided preconditioning, first compute $\widehat{W} = \widehat{B}_L$ and let $W = P_L A$. Then compute

$$B_{LR} = P_L A P_R, \quad \widehat{B}_{LR} = \mathfrak{fl}(\widehat{W} \widehat{P}_R).$$

Let

$$\|A\|_{1,1} = \sum_{p=1}^m \sum_{q=1}^n |A_{pq}|.$$

Then the row-sensitive bound is

$$|(\widehat{B}_{LR})_{ij} - (B_{LR})_{ij}| \leq (\gamma_n \widehat{p}_n + \alpha_n) \|W_{i*}\|_1 + (1 + \gamma_n) \widehat{p}_n \kappa_m \|A\|_{1,1}.$$

Consequently,

$$|(\widehat{B}_{LR})_{ij} - (B_{LR})_{ij}| \leq (\gamma_n \widehat{p}_n + \alpha_n) \sqrt{n} \|A\|_F + (1 + \gamma_n) \widehat{p}_n \kappa_m \|A\|_{1,1}.$$

These three displayed bounds are the Algorithm 3 floating-point changes for the computed matrices $\Pi_L A$, $A \Pi_R$, and $\Pi_L A \Pi_R$ on the materialized scaled-pattern SRHT path.

Theorem 14 (Computed materialized SRHT basis and uniform-row Gram). *Let $U \in \mathbb{R}^{m \times n}$ be the exact orthonormal-column analysis basis used in the SRHT subspace-embedding theorem. Let $S \in \mathbb{R}^{m \times m}$ be an exact sign-pattern table with $S_{ia}^2 = 1$, and let $\omega \in \{\pm 1\}^m$ be the exact realized Rademacher sign vector. Store the signs by one of the rounded copy routines $\mathfrak{fl}(\omega_a \cdot 1)$, $\mathfrak{fl}(\omega_a + 0)$, or $\mathfrak{fl}(\omega_a - 0)$, and write the stored signs as $\widehat{\omega}^\rho$. Then*

$$|\widehat{\omega}_a^\rho - \omega_a| \leq u.$$

Define

$$c_m = \sqrt{1/m}, \quad \widehat{c}_m = \mathfrak{fl}(\sqrt{1/m}), \\ H_{ia} = c_m S_{ia}, \quad \widehat{H}_{ia} = \widehat{c}_m S_{ia}, \quad E_{ia}^H = c_m u |S_{ia}|.$$

Set

$$D_\omega = \text{diag}(\omega), \quad \widehat{D}_\omega^\rho = \text{diag}(\widehat{\omega}^\rho), \\ P_\omega = H D_\omega, \quad \widehat{P}_\omega^\rho = \mathfrak{fl}(\widehat{H} \widehat{D}_\omega^\rho).$$

Define

$$V_\omega = P_\omega U, \quad \widehat{V}_\omega^\rho = \mathfrak{fl}(\widehat{P}_\omega^\rho U).$$

Define the computed-basis entry radius

$$\Delta_V = \kappa_m \sqrt{m} = \gamma_m (2 + \gamma_m) (1 + u)^2 + u(1 + u) + u.$$

Then

$$|\widehat{V}_{\omega,ij}^\rho - (V_\omega)_{ij}| \leq \Delta_V.$$

The scalar Δ_V uses only the universal column bound $\|U_{*j}\|_1 \leq \sqrt{m}$, which follows from $U^T U = I_n$. Thus no column-by-column property of U is needed in the final displayed radius.

Let $J = (J_1, \dots, J_s)$ be the exact uniform row sample. The final concrete SRHT endpoint uses the computed denominator

$$d = \sqrt{s/m}, \quad \widehat{d}_\star = \mathfrak{fl}(\mathfrak{fl}(\sqrt{s}) \mathfrak{fl}(1/\mathfrak{fl}(\sqrt{m}))),$$

and the locally proved denominator certificate gives

$$\widehat{d}_\star \neq 0, \quad |\widehat{d}_\star - d| \leq d \frac{4u + 3u^2 + u^3}{1 - u}.$$

Define the concrete relative denominator error

$$\delta_d = \frac{4u + 3u^2 + u^3}{1 - u}, \quad 0 \leq \delta_d < 1,$$

and the relative scaled-row entry error

$$\zeta(t) = \frac{S(t)(u + \delta_d) + (1 + u)\Delta_V}{(1 - \delta_d)S(t)}.$$

On the exact SRHT row-norm event, every exact scaled sketch entry has absolute value at most $\sqrt{m/s} S(t)$, and every computed scaled sketch entry differs from it by at most $\sqrt{m/s} S(t)\zeta(t)$.

Let the computed sampled matrix be

$$\widehat{B}_{tj}^\rho = \text{fl}\left(\widehat{V}_{\omega, J_{tj}}^\rho / \widehat{d}_\star\right), \quad \widehat{G}_{\omega, J}^\rho = \text{fl}\left((\widehat{B}^\rho)^T \widehat{B}^\rho\right).$$

Define the final readable floating-point radius

$$T_{\text{SRHT}}^{\text{fp}}(t) = nm S(t)^2 \left(2\zeta(t) + \zeta(t)^2 + \gamma_s(1 + \zeta(t))^2\right).$$

Equivalently, substituting $\zeta(t)$, $S(t)$, δ_d , and Δ_V , the same exact radius is

$$\begin{aligned} T_{\text{SRHT}}^{\text{fp}}(t) = nm \left(\sqrt{\frac{n}{m}} + t \right)^2 & \left[2 \frac{\left(\sqrt{\frac{n}{m}} + t \right) \left(u + \frac{4u + 3u^2 + u^3}{1 - u} \right) + (1 + u) \left(\gamma_m(2 + \gamma_m)(1 + u)^2 + u(1 + u) + u \right)}{\left(1 - \frac{4u + 3u^2 + u^3}{1 - u} \right) \left(\sqrt{\frac{n}{m}} + t \right)} \right. \\ & + \left(\frac{\left(\sqrt{\frac{n}{m}} + t \right) \left(u + \frac{4u + 3u^2 + u^3}{1 - u} \right) + (1 + u) \left(\gamma_m(2 + \gamma_m)(1 + u)^2 + u(1 + u) + u \right)}{\left(1 - \frac{4u + 3u^2 + u^3}{1 - u} \right) \left(\sqrt{\frac{n}{m}} + t \right)} \right)^2 \\ & \left. + \gamma_s \left(1 + \frac{\left(\sqrt{\frac{n}{m}} + t \right) \left(u + \frac{4u + 3u^2 + u^3}{1 - u} \right) + (1 + u) \left(\gamma_m(2 + \gamma_m)(1 + u)^2 + u(1 + u) + u \right)}{\left(1 - \frac{4u + 3u^2 + u^3}{1 - u} \right) \left(\sqrt{\frac{n}{m}} + t \right)} \right)^2 \right]. \end{aligned}$$

In the small-roundoff regime

$$0 \leq u, \gamma_m, \gamma_s, \frac{4u + 3u^2 + u^3}{1 - u} \leq c < 1, \quad \sqrt{\frac{n}{m}} + t > 0,$$

the component terms have the orders

$$\Delta_V = \Theta(\gamma_m + u), \quad \zeta(t) = \Theta\left(u + \frac{4u + 3u^2 + u^3}{1 - u} + \frac{\gamma_m + u}{\sqrt{n/m} + t}\right),$$

with constants depending only on the fixed bound c . Since all terms are nonnegative and

$$2\zeta + \zeta^2 + \gamma_s(1 + \zeta)^2 = \Theta(\zeta + \gamma_s) \quad (0 \leq \zeta, \gamma_s \leq c),$$

the final radius has the asymptotic order

$$T_{\text{SRHT}}^{\text{fp}}(t) = \Theta \left(nm \left(\sqrt{\frac{n}{m}} + t \right)^2 \left[\gamma_s + u + \frac{4u + 3u^2 + u^3}{1 - u} + \frac{\gamma_m + u}{\sqrt{n/m} + t} \right] \right).$$

Equivalently,

$$T_{\text{SRHT}}^{\text{fp}}(t) = \Theta \left(nm \left(\sqrt{\frac{n}{m}} + t \right)^2 \left(\gamma_s + u + \frac{4u + 3u^2 + u^3}{1 - u} \right) + nm \left(\sqrt{\frac{n}{m}} + t \right) (\gamma_m + u) \right).$$

In the same irreducible variables,

$$T_{\text{SRHT}}^{\text{fp}}(t) = \Theta \left(nm \left(\sqrt{\frac{n}{m}} + t \right)^2 \left(\gamma_s + u + \frac{4u + 3u^2 + u^3}{1 - u} \right) + nm \left(\sqrt{\frac{n}{m}} + t \right) (\gamma_m + u) \right).$$

Assume the exact SRHT preprocessing and uniform-row sample budgets

$$m \exp \left(-\frac{mt^2}{8} \right) \leq \delta_{\text{pre}}, \quad \Phi(\theta, L(t), \varepsilon, s, n) \leq \delta_{\text{sample}}.$$

Then the fully computed materialized SRHT Gram satisfies

$$\mathbb{P} \left[-(\varepsilon + T_{\text{SRHT}}^{\text{fp}}(t))I_n \preceq \widehat{G}_{\omega, J}^p - I_n \preceq (\varepsilon + T_{\text{SRHT}}^{\text{fp}}(t))I_n \right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

No additional perturbation event is used. The radius depends only on $u, \gamma_m, \gamma_s, m, n, s$, the SRHT parameter t , and the concrete relative denominator error $\delta_d = (4u + 3u^2 + u^3)/(1 - u)$. For fixed m, n, s, t ,

$$T_{\text{SRHT}}^{\text{fp}}(t) \longrightarrow 0 \quad \text{as} \quad u, \gamma_m, \gamma_s, \frac{4u + 3u^2 + u^3}{1 - u} \longrightarrow 0,$$

so the bound is non-vacuous in the small-roundoff regime. In the exact-denominator infrastructure specialization, $\delta_d = 0$; that is not the selected final concrete denominator routine in this note.

Remark. The previous theorem is an analysis-basis theorem. It is the right statement when the exact SRHT subspace-embedding analysis is written for an orthonormal-column matrix U , but it is not by itself the Algorithm 3 input-matrix theorem. Algorithm 3 receives A , and the computed product is $\text{fl}(\widehat{P}_\omega A)$, not $\text{fl}(\widehat{P}_\omega U)$.

Theorem 15 (Fully computed SRHT theorem on the Algorithm 3 input matrix). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix handed to Algorithm 3. Assume $A \neq 0$, let $r = \text{rank}(A)$, and choose exact analysis-only factors*

$$U_A \in \mathbb{R}^{m \times r}, \quad C_A \in \mathbb{R}^{r \times n}, \quad U_A^T U_A = I_r, \quad A = U_A C_A.$$

The matrices U_A and C_A are exact arithmetic objects used only in the analysis. They are not computed by Algorithm 3. The implemented algorithm computes with the input matrix A .

Let $0 < \delta_{\text{pre}} < m$, $0 \leq \delta_{\text{sample}}$, and $0 < \varepsilon < 1$. Set

$$t_{\text{pre}} = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}, \quad S_A = \sqrt{\frac{r}{m}} + t_{\text{pre}}, \quad L_A = m S_A^2.$$

Choose the exact sample count s and an exact Chernoff parameter $\theta_A > 0$ so that

$$\Phi(\theta_A, L_A, \varepsilon, s, r) \leq \delta_{\text{sample}}.$$

The choice of s , θ_A , and the probability laws below is exact arithmetic; no sampling probability table is computed.

Let $\mathcal{S} \in \mathbb{R}^{m \times m}$ be an exact sign-pattern table such that

$$H_{ia} = \sqrt{1/m} \mathcal{S}_{ia}$$

is orthogonal and flat. Let $\omega \in \{\pm 1\}^m$ be the exact Rademacher sign vector, and let $J = (J_1, \dots, J_s)$ be the exact uniform row-sample trace. The concrete computed SRHT path in this theorem is

$$\begin{aligned} \hat{H}_{ia} &= \text{fl}(\sqrt{1/m}) \mathcal{S}_{ia}, & \hat{\omega}_a &= \text{fl}(\omega_a \cdot 1), & \hat{D}_\omega &= \text{diag}(\hat{\omega}), \\ P_\omega &= H \text{diag}(\omega), & \hat{P}_\omega &= \text{fl}(\hat{H} \hat{D}_\omega), & Y_\omega &= P_\omega A, & \hat{Y}_\omega &= \text{fl}(\hat{P}_\omega A). \end{aligned}$$

Here P_ω, Y_ω are exact references, while $\hat{P}_\omega, \hat{Y}_\omega$ are computed floating-point objects. The stored-sign path displayed here uses $\text{fl}(\omega_a \cdot 1)$; the formalization also has the analogous add-zero and subtract-zero stored-sign certificates.

Let the final concrete uniform row-scale denominator be

$$d = \sqrt{s/m}, \quad \hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{m}))).$$

The value $\hat{d}_\star$ is a computed non-probability quantity, and the formalized certificate proves

$$\hat{d}_\star \neq 0, \quad |\hat{d}_\star - d| \leq d \frac{4u + 3u^2 + u^3}{1 - u}.$$

Define

$$B_{\omega, J, tj} = \frac{(Y_\omega)_{J_t j}}{d}, \quad \tilde{B}_{\omega, J, tj} = \frac{(\hat{Y}_\omega)_{J_t j}}{d}, \quad \hat{B}_{\omega, J, tj} = \text{fl}\left(\frac{(\hat{Y}_\omega)_{J_t j}}{\hat{d}_\star}\right),$$

and

$$\hat{G}_A(\omega, J) = \text{fl}\left(\hat{B}_{\omega, J}^T \hat{B}_{\omega, J}\right).$$

The fully computed perturbation radius is

$$\tau_A^{\text{cd}}(\omega, J) = \tau_{\text{den}}(\omega, J) + \tau_{\text{dot}}(\omega, J) + \tau_{\text{prod}}(\omega, J).$$

First define the row-scaling entry budget

$$E_{tj}^{\text{den}} = |\hat{B}_{\omega, J, tj}| u + \frac{|(\hat{Y}_\omega)_{J_t j}| d \frac{4u + 3u^2 + u^3}{1 - u}}{|\hat{d}_\star| d}.$$

Then

$$\tau_{\text{den}}(\omega, J) = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den}} |\hat{B}_{\omega, J, tk}| + |\tilde{B}_{\omega, J, tj}| E_{tk}^{\text{den}} \right)_{j,k=1}^n \right\|_F$$

and

$$\tau_{\text{dot}}(\omega, J) = \left\| \left(\gamma_s \sum_{t=1}^s |\hat{B}_{\omega, J, tj}| |\hat{B}_{\omega, J, tk}| \right)_{j,k=1}^n \right\|_F.$$

For the product term, let

$$e_{ia}^H = \sqrt{1/m} u |\mathcal{S}_{ia}|, \quad e_a^\omega = u,$$

and define the preconditioner-entry budget

$$E_{\omega,ia}^P = \gamma_m \sum_{b=1}^m |\hat{H}_{ib}| |\hat{D}_{\omega,ba}| + \sum_{b=1}^m e_{ib}^H |\hat{D}_{\omega,ba}| + \sum_{b=1}^m |H_{ib}| e_b^\omega \mathbf{1}_{b=a}.$$

The entrywise budget for the computed product $\hat{Y}_\omega = \text{fl}(\hat{P}_\omega A)$ is

$$E_{\omega,ij}^Y = \gamma_m \sum_{a=1}^m |(\hat{P}_\omega)_{ia}| |A_{aj}| + \sum_{a=1}^m E_{\omega,ia}^P |A_{aj}|,$$

and

$$\tau_{\text{prod}}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega,J,tj}^Y}{d} |\tilde{B}_{\omega,J,tk}| + |B_{\omega,J,tj}| \frac{E_{\omega,J,tk}^Y}{d} \right)_{j,k=1}^n \right\|_F.$$

Then the fully computed Algorithm 3 sampled input Gram satisfies

$$\mathbb{P} \left[(1 - \varepsilon) A^T A - \tau_A^{\text{cd}}(\omega, J) I_n \preceq \hat{G}_A(\omega, J) \preceq (1 + \varepsilon) A^T A + \tau_A^{\text{cd}}(\omega, J) I_n \right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

There is no extra perturbation event, no extra existence hypothesis, and no δ_{comp} in this final statement. All probabilities are exact Rademacher and exact uniform-row probabilities. All non-probability computed objects on this SRHT path are charged in τ_A^{cd} .

Theorem 16 (Concrete Sylvester/Walsh FHT specialization). *Fix $p \in \mathbb{N}$ and set $m = 2^p$. Define the exact Sylvester/Walsh sign table*

$$S_p(i, a) = \begin{cases} -1, & \text{if the bit parity of } i \wedge a \text{ is odd,} \\ 1, & \text{otherwise,} \end{cases}$$

and define the normalized exact table

$$H_p(i, a) = \sqrt{1/m} S_p(i, a).$$

The formalized result proves from this definition that $H_p^T H_p = I_m$ and $(H_p)_{ia}^2 = 1/m$. Thus these are exact arithmetic facts about the generated bit table, not hypotheses supplied to the final theorem.

Let the exact Algorithm 3 input have the analysis factorization

$$A = U_A C_A, \quad U_A^T U_A = I_r.$$

The matrices U_A and C_A are exact analysis-only witnesses. The implemented computation uses A directly. For an exact Rademacher sign vector ω , the concrete computed preconditioner is obtained by storing the realized signs with the rounded multiply-one copy path, applying the generated Sylvester/Walsh FHT butterfly schedule, and rounding the normalization by $\text{fl}(\sqrt{1/m})$. It then computes

$$\hat{Y}_\omega = \text{fl}(\hat{P}_\omega A).$$

For the downstream uniform row sketch, the concrete computed denominator is

$$d = \sqrt{s/m}, \quad \hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{m}))),$$

with the same certified radius

$$|\widehat{d}_\star - d| \leq d \frac{4u + 3u^2 + u^3}{1 - u}.$$

Let $\tau_{A,\text{syl}}^{\text{cd}}(\omega, J)$ be the fully expanded radius $\tau_A^{\text{cd}}(\omega, J)$ from the preceding theorem after substituting $\mathcal{S} = S_p$, $H = H_p$, $m = 2^p$, and the generated-FHT stored-sign preconditioner just described.

With

$$t_{\text{pre}} = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}, \quad S_A = \sqrt{\frac{r}{m}} + t_{\text{pre}}, \quad L_A = mS_A^2,$$

and with the same exact sample-budget condition

$$\Phi(\theta_A, L_A, \varepsilon, s, r) \leq \delta_{\text{sample}},$$

the computed sampled input Gram satisfies

$$\mathbb{P}\left[(1 - \varepsilon)A^T A - \tau_{A,\text{syl}}^{\text{cd}}(\omega, J)I_n \preceq \widehat{G}_A(\omega, J) \preceq (1 + \varepsilon)A^T A + \tau_{A,\text{syl}}^{\text{cd}}(\omega, J)I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

The exact objects are A , U_A , C_A , S_p , H_p , and the Rademacher and uniform-row laws. The computed objects are the stored sign vector, generated FHT butterflies, rounded normalization, $\widehat{P}_\omega$, $\widehat{Y}_\omega$, $\widehat{d}_\star$, sampled row divisions, and Gram dot products. No external orthogonality witness, flatness witness, perturbation event, or extra existence hypothesis appears.

Corollary 17 (Stored-input generated-FHT endpoints). *Keep the hypotheses and source sample-budget condition of the preceding Sylvester/Walsh specialization. Let*

$$P_\omega = H_p \text{diag}(\omega), \quad Y_\omega = P_\omega A, \quad d = \sqrt{s/m}.$$

These are exact analysis references. The exact matrices U_A , C_A , $A = U_A C_A$, H_p , and P_ω are not computed in this corollary, and the Rademacher and uniform-row laws are exact mathematical laws.

Choose one concrete input-storage mode

$$\rho \in \{\text{mul1}, \text{add0}, \text{sub0}\}.$$

The implemented input-storage step is

$$\widehat{A}_{aj}^{\text{in}} = \begin{cases} \text{fl}_{\text{mul}}(A_{aj}, 1), & \rho = \text{mul1}, \\ \text{fl}_{\text{add}}(A_{aj}, 0), & \rho = \text{add0}, \\ \text{fl}_{\text{sub}}(A_{aj}, 0), & \rho = \text{sub0}, \end{cases} \quad E_{aj}^A = u|A_{aj}|,$$

so

$$|\widehat{A}_{aj}^{\text{in}} - A_{aj}| \leq E_{aj}^A.$$

This is the only new computed object relative to the preceding generated-FHT endpoint. The stored-sign generated-FHT preconditioner $\widehat{P}_\omega$, its exact reference P_ω , and its entrywise budget $E_{\omega,ia}^P$ are the same objects used above:

$$|(\widehat{P}_\omega)_{ia} - (P_\omega)_{ia}| \leq E_{\omega,ia}^P.$$

The implementation then forms the computed preconditioned input

$$\widehat{Y}_\omega^{\text{in}} = \text{fl}(\widehat{P}_\omega \widehat{A}^{\text{in}}).$$

The fully expanded entrywise product budget is

$$E_{\omega,ij}^{Y,\text{in}} = \gamma_m \sum_{a=1}^m |(\hat{P}_\omega)_{ia}| |\hat{A}_{aj}^{\text{in}}| + \sum_{a=1}^m E_{\omega,ia}^P |\hat{A}_{aj}^{\text{in}}| + \sum_{a=1}^m |(P_\omega)_{ia}| u |A_{aj}|.$$

The last summand is the exact input-storage contribution; it is not a probability term and not a hidden certificate.

Use the same computed denominator

$$\hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{m}))) , \quad |\hat{d}_\star - d| \leq d \delta_d, \quad \delta_d = \frac{4u + 3u^2 + u^3}{1 - u}.$$

For sampled rows $J_1, \dots, J_s$, define

$$B_{\omega,J,tj} = \frac{(Y_\omega)_{J_tj}}{d}, \quad \tilde{B}_{\omega,J,tj}^{\text{in}} = \frac{(\hat{Y}_\omega^{\text{in}})_{J_tj}}{d}, \quad \hat{B}_{\omega,J,tj}^{\text{in}} = \text{fl}\left(\frac{(\hat{Y}_\omega^{\text{in}})_{J_tj}}{\hat{d}_\star}\right).$$

The computed sampled Gram is

$$\hat{G}_A^{\text{in}}(\omega, J) = \text{fl}\left((\hat{B}_{\omega,J}^{\text{in}})^T \hat{B}_{\omega,J}^{\text{in}}\right).$$

The row-scaling budget is

$$E_{tj}^{\text{den},\text{in}} = |\hat{B}_{\omega,J,tj}^{\text{in}}| u + \frac{|(\hat{Y}_\omega^{\text{in}})_{J_tj}| d \delta_d}{|\hat{d}_\star| d}.$$

Define

$$\tau_{\text{den},\text{in}}(\omega, J) = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den},\text{in}} |\hat{B}_{\omega,J,tj}^{\text{in}}| + |\tilde{B}_{\omega,J,tj}^{\text{in}}| E_{tk}^{\text{den},\text{in}} \right)_{j,k=1}^n \right\|_F,$$

$$\tau_{\text{dot},\text{in}}(\omega, J) = \left\| \left(\gamma_s \sum_{t=1}^s |\hat{B}_{\omega,J,tj}^{\text{in}}| |\hat{B}_{\omega,J,tk}^{\text{in}}| \right)_{j,k=1}^n \right\|_F,$$

and

$$\tau_{\text{prod},\text{in}}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega,J,tj}^{Y,\text{in}}}{d} |\tilde{B}_{\omega,J,tk}^{\text{in}}| + |B_{\omega,J,tj}| \frac{E_{\omega,J,tk}^{Y,\text{in}}}{d} \right)_{j,k=1}^n \right\|_F.$$

The final stored-input radius is

$$\tau_{A,\text{in}}^{\text{cd}}(\omega, J) = \tau_{\text{den},\text{in}}(\omega, J) + \tau_{\text{dot},\text{in}}(\omega, J) + \tau_{\text{prod},\text{in}}(\omega, J).$$

Then

$$\mathbb{P}\left[(1 - \varepsilon)A^T A - \tau_{A,\text{in}}^{\text{cd}}(\omega, J)I_n \preceq \hat{G}_A^{\text{in}}(\omega, J) \preceq (1 + \varepsilon)A^T A + \tau_{A,\text{in}}^{\text{cd}}(\omega, J)I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

The input-storage part of the product radius is exactly

$$\Delta_{\text{in}}(\omega, J) = u \left\| \left(\sum_{t=1}^s \frac{\sum_{a=1}^m |(P_\omega)_{J_t a}| |A_{aj}|}{d} |\tilde{B}_{\omega,J,tk}^{\text{in}}| + |B_{\omega,J,tj}| \frac{\sum_{a=1}^m |(P_\omega)_{J_t a}| |A_{ak}|}{d} \right)_{j,k=1}^n \right\|_F.$$

Thus, for fixed m, n, s, p, A, ω, J , fixed $d > 0$, and nonzero displayed coefficient,

$$\Delta_{\text{in}}(\omega, J) = \Theta \left(u \left\| \left(\sum_{t=1}^s \frac{\sum_{a=1}^m |(P_\omega)_{Jta}| |A_{aj}|}{d} |\tilde{B}_{\omega, J, tk}^{\text{in}}| + |B_{\omega, J, tj}| \frac{\sum_{a=1}^m |(P_\omega)_{Jta}| |A_{ak}|}{d} \right)_{j,k=1}^n \right\|_F \right).$$

The whole radius has the same non-vacuity property as the preceding generated-FHT endpoint: for fixed dimensions, fixed exact references, and sample realizations with $\hat{d}_\star$ bounded away from zero,

$$\tau_{A, \text{in}}^{\text{cd}}(\omega, J) \rightarrow 0 \quad \text{as} \quad u, \gamma_m, \gamma_s, E^P \rightarrow 0.$$

Corollary 18 (All-coordinate add-zero FHT writeback specialization). *Keep the notation and hypotheses of the preceding Sylvester/Walsh specialization. Define the exact signed-Hadamard preconditioner*

$$P_\omega = H_p \text{diag}(\omega).$$

Let $\mathcal{P}_p = ((q_\ell, r_\ell))_{\ell=1}^{L_p}$ be the deterministic ordered pair schedule used by the generated Sylvester/Walsh FHT. Here $q_\ell, r_\ell \in \{1, \dots, m\}$, and L_p is the number of butterfly pairs in the schedule. For the standard radix-two Sylvester schedule, $L_p = p2^{p-1}$ when $p > 0$, and $L_p = 0$ when $p = 0$.

For each column a of the diagonal sign matrix, initialize the exact FHT input, computed stored input, and entrywise input-error budget by

$$x_i^{(0,a)} = \omega_a \mathbf{1}_{i=a}, \quad \hat{x}_i^{(0,a)} = \begin{cases} \text{fl}(\omega_a \cdot 1), & i = a, \\ 0, & i \neq a, \end{cases} \quad E_i^{(0,a)} = u \mathbf{1}_{i=a}.$$

For $\ell = 1, \dots, L_p$, write $q = q_\ell$ and $r = r_\ell$. The exact reference vector $x^{(\ell,a)}$ is obtained from $x^{(\ell-1,a)}$ by the exact butterfly $x_q + x_r$, $x_q - x_r$, leaving all other coordinates unchanged. The computed pre-writeback vector $y^{(\ell,a)}$ is

$$y_q^{(\ell,a)} = \text{fl}(\hat{x}_q^{(\ell-1,a)} + \hat{x}_r^{(\ell-1,a)}), \quad y_r^{(\ell,a)} = \text{fl}(\hat{x}_q^{(\ell-1,a)} - \hat{x}_r^{(\ell-1,a)}),$$

and $y_i^{(\ell,a)} = \hat{x}_i^{(\ell-1,a)}$ for $i \notin \{q, r\}$. Before the explicit copy/writeback, define

$$F_i^{(\ell,a)} = \begin{cases} u|\hat{x}_q^{(\ell-1,a)} + \hat{x}_r^{(\ell-1,a)}| + E_q^{(\ell-1,a)} + E_r^{(\ell-1,a)}, & i = q, \\ u|\hat{x}_q^{(\ell-1,a)} - \hat{x}_r^{(\ell-1,a)}| + E_q^{(\ell-1,a)} + E_r^{(\ell-1,a)}, & i = r, \\ E_i^{(\ell-1,a)}, & i \notin \{q, r\}. \end{cases}$$

The all-coordinate add-zero implementation stores every post-butterfly coordinate, including untouched coordinates, by

$$\hat{x}_i^{(\ell,a)} = \text{fl}(y_i^{(\ell,a)} + 0), \quad E_i^{(\ell,a)} = F_i^{(\ell,a)} + u|y_i^{(\ell,a)}|.$$

The final term $u|y_i^{(\ell,a)}|$ is precisely the additional floating-point writeback/copy charge. It is a computed non-probability operation; it does not alter the Rademacher or row-sampling law.

Set

$$c_p = \sqrt{1/m}, \quad \hat{c}_p = \text{fl}(\sqrt{1/m}), \quad \eta_p = c_p u.$$

The computed add-zero FHT preconditioner and its entrywise error budget are

$$\hat{P}_{\omega, ia}^+ = \text{fl}(\hat{c}_p \hat{x}_i^{(L_p, a)}),$$

and

$$E_{\omega,ia}^{P,+} = u|\widehat{c}_p \widehat{x}_i^{(L_p,a)}| + |\widehat{c}_p| E_i^{(L_p,a)} + \eta_p(|\widehat{x}_i^{(L_p,a)}| + E_i^{(L_p,a)}).$$

Thus

$$|\widehat{P}_{\omega,ia}^+ - (P_\omega)_{ia}| \leq E_{\omega,ia}^{P,+}.$$

Define the computed left-preconditioned matrix

$$\widehat{Y}_\omega^+ = \mathfrak{fl}(\widehat{P}_\omega^+ A)$$

and the corresponding product budget

$$E_{\omega,ij}^{Y,+} = \gamma_m \sum_{a=1}^m |\widehat{P}_{\omega,ia}^+| |A_{aj}| + \sum_{a=1}^m E_{\omega,ia}^{P,+} |A_{aj}|.$$

With the same exact $d = \sqrt{s/m}$ and the same computed $\widehat{d}_\star$, set

$$B_{\omega,J,tj} = \frac{(Y_\omega)_{J,tj}}{d}, \quad \widetilde{B}_{\omega,J,tj}^+ = \frac{(\widehat{Y}_\omega^+)_{J,tj}}{d}, \quad \widehat{B}_{\omega,J,tj}^+ = \mathfrak{fl}\left(\frac{(\widehat{Y}_\omega^+)_{J,tj}}{\widehat{d}_\star}\right),$$

and

$$\widehat{G}_A^+(\omega, J) = \mathfrak{fl}\left((\widehat{B}_{\omega,J}^+)^T \widehat{B}_{\omega,J}^+\right).$$

Let

$$\delta_d = \frac{4u + 3u^2 + u^3}{1 - u}.$$

The row-scaling budget is

$$E_{tj}^{\text{den},+} = |\widehat{B}_{\omega,J,tj}^+| u + \frac{|(\widehat{Y}_\omega^+)_{J,tj}| d \delta_d}{|\widehat{d}_\star| d}.$$

Define

$$\tau_{\text{den},+}(\omega, J) = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den},+} |\widehat{B}_{\omega,J,tk}^+| + |\widetilde{B}_{\omega,J,tj}^+| E_{tk}^{\text{den},+} \right)_{j,k=1}^n \right\|_F,$$

$$\tau_{\text{dot},+}(\omega, J) = \left\| \left(\gamma_s \sum_{t=1}^s |\widehat{B}_{\omega,J,tj}^+| |\widehat{B}_{\omega,J,tk}^+| \right)_{j,k=1}^n \right\|_F,$$

and

$$\tau_{\text{prod},+}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega,J,tj}^{Y,+}}{d} |\widetilde{B}_{\omega,J,tk}^+| + |B_{\omega,J,tj}| \frac{E_{\omega,J,tk}^{Y,+}}{d} \right)_{j,k=1}^n \right\|_F.$$

Finally set

$$\tau_{A,\text{syl},+}^{\text{cd}}(\omega, J) = \tau_{\text{den},+}(\omega, J) + \tau_{\text{dot},+}(\omega, J) + \tau_{\text{prod},+}(\omega, J).$$

Under the same exact preprocessing and sample-budget condition as in the preceding theorem,

$$\mathbb{P}\left[(1 - \varepsilon)A^T A - \tau_{A,\text{syl},+}^{\text{cd}}(\omega, J)I_n \preceq \widehat{G}_A^+(\omega, J) \preceq (1 + \varepsilon)A^T A + \tau_{A,\text{syl},+}^{\text{cd}}(\omega, J)I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

For fixed p, m, n, s, A, ω, J , the additional writeback terms are finite sums of factors proportional to u . In the small-roundoff regime $\gamma_m = \Theta(mu)$, $\gamma_s = \Theta(su)$, and $\delta_d = \Theta(u)$, the displayed deterministic radius satisfies

$$\tau_{A, \text{syl}, +}^{\text{cd}}(\omega, J) = O\left((u + \gamma_m + \gamma_s + \delta_d) \left[\sum_{i,a} |\hat{x}_i^{(L_p, a)}| \|A_{a,*}\|_1 + \sum_{\ell=1}^{L_p} \sum_{i,a} |y_i^{(\ell, a)}| \|A_{a,*}\|_1 + \sum_{t,j,k} |\hat{B}_{\omega, J, tj}^+| |\hat{B}_{\omega, J, tk}^+| \right] \right),$$

with Θ of the same expanded expression whenever the bracketed realized magnitude is bounded above and below by positive constants independent of u . In particular, $\tau_{A, \text{syl}, +}^{\text{cd}}(\omega, J) \rightarrow 0$ as the roundoff quantities tend to zero with the dimensions, exact input, and realized randomness fixed.

Corollary 19 (Modified-coordinate add-zero FHT writeback specialization). *Keep the notation and hypotheses of Corollary 17. The modified-coordinate writeback implementation uses the same exact P_ω , the same pair schedule $\mathcal{P}_p = ((q_\ell, r_\ell))_{\ell=1}^{L_p}$, and the same computed pre-writeback vector $y^{(\ell, a)}$ and pre-writeback budget $F^{(\ell, a)}$. It differs only in the storage step after each butterfly: for $q = q_\ell$ and $r = r_\ell$,*

$$\hat{x}_i^{(\ell, a), \text{mc}} = \begin{cases} \text{fl}(y_i^{(\ell, a), \text{mc}} + 0), & i \in \{q, r\}, \\ y_i^{(\ell, a), \text{mc}}, & i \notin \{q, r\}, \end{cases}$$

and

$$E_i^{(\ell, a), \text{mc}} = \begin{cases} F_i^{(\ell, a), \text{mc}} + u|y_i^{(\ell, a), \text{mc}}|, & i \in \{q, r\}, \\ F_i^{(\ell, a), \text{mc}}, & i \notin \{q, r\}. \end{cases}$$

The superscript **mc** denotes the modified-coordinate recurrence. Thus only the two coordinates whose FHT values are changed by the butterfly receive the extra add-zero copy charge.

With the same

$$c_p = \sqrt{1/m}, \quad \hat{c}_p = \text{fl}(\sqrt{1/m}), \quad \eta_p = c_p u,$$

define

$$\hat{P}_{\omega, ia}^{\text{mc}} = \text{fl}(\hat{c}_p \hat{x}_i^{(L_p, a), \text{mc}})$$

and

$$E_{\omega, ia}^{P, \text{mc}} = u|\hat{c}_p \hat{x}_i^{(L_p, a), \text{mc}}| + |\hat{c}_p| E_i^{(L_p, a), \text{mc}} + \eta_p (|\hat{x}_i^{(L_p, a), \text{mc}}| + E_i^{(L_p, a), \text{mc}}).$$

Then

$$|\hat{P}_{\omega, ia}^{\text{mc}} - (P_\omega)_{ia}| \leq E_{\omega, ia}^{P, \text{mc}}.$$

Define

$$\begin{aligned} \hat{Y}_\omega^{\text{mc}} &= \text{fl}(\hat{P}_\omega^{\text{mc}} A), \\ E_{\omega, ij}^{Y, \text{mc}} &= \gamma_m \sum_{a=1}^m |\hat{P}_{\omega, ia}^{\text{mc}}| |A_{aj}| + \sum_{a=1}^m E_{\omega, ia}^{P, \text{mc}} |A_{aj}|. \end{aligned}$$

Using the same concrete denominator $\hat{d}_\star$, define

$$\tilde{B}_{\omega, J, tj}^{\text{mc}} = \frac{(\hat{Y}_\omega^{\text{mc}})_{J, tj}}{d}, \quad \hat{B}_{\omega, J, tj}^{\text{mc}} = \text{fl}\left(\frac{(\hat{Y}_\omega^{\text{mc}})_{J, tj}}{\hat{d}_\star}\right),$$

and

$$\widehat{G}_A^{\text{mc}}(\omega, J) = \text{fl} \left((\widehat{B}_{\omega, J}^{\text{mc}})^T \widehat{B}_{\omega, J}^{\text{mc}} \right).$$

Define

$$E_{tj}^{\text{den,mc}} = |\widehat{B}_{\omega, J, tj}^{\text{mc}}| u + \frac{|(\widehat{Y}_{\omega}^{\text{mc}})_{J, tj}| d \delta_d}{|\widehat{d}_{\star}| d}.$$

Then

$$\tau_{\text{den,mc}}(\omega, J) = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den,mc}} |\widehat{B}_{\omega, J, tk}^{\text{mc}}| + |\widetilde{B}_{\omega, J, tj}^{\text{mc}}| E_{tk}^{\text{den,mc}} \right)_{j,k=1}^n \right\|_F,$$

$$\tau_{\text{dot,mc}}(\omega, J) = \left\| \left(\gamma_s \sum_{t=1}^s |\widehat{B}_{\omega, J, tj}^{\text{mc}}| |\widehat{B}_{\omega, J, tk}^{\text{mc}}| \right)_{j,k=1}^n \right\|_F,$$

and

$$\tau_{\text{prod,mc}}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega, J, tj}^{Y, \text{mc}}}{d} |\widetilde{B}_{\omega, J, tk}^{\text{mc}}| + |B_{\omega, J, tj}| \frac{E_{\omega, J, tk}^{Y, \text{mc}}}{d} \right)_{j,k=1}^n \right\|_F.$$

The full modified-coordinate radius is

$$\tau_{A, \text{sy1,mc}}^{\text{cd}}(\omega, J) = \tau_{\text{den,mc}}(\omega, J) + \tau_{\text{dot,mc}}(\omega, J) + \tau_{\text{prod,mc}}(\omega, J).$$

Under the same exact preprocessing and sample-budget condition,

$$\mathbb{P} \left[(1 - \varepsilon) A^T A - \tau_{A, \text{sy1,mc}}^{\text{cd}}(\omega, J) I_n \preceq \widehat{G}_A^{\text{mc}}(\omega, J) \preceq (1 + \varepsilon) A^T A + \tau_{A, \text{sy1,mc}}^{\text{cd}}(\omega, J) I_n \right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

For fixed p, m, n, s, A, ω, J , the extra modified-coordinate copy terms are finite sums of factors $u |y_i^{(\ell, a), \text{mc}}|$ only over $i \in \{q_\ell, r_\ell\}$. In the same small-roundoff regime,

$$\tau_{A, \text{sy1,mc}}^{\text{cd}}(\omega, J) = O \left(\tau_{A, \text{sy1},+}^{\text{cd}}(\omega, J) \right),$$

and its irreducible copy contribution replaces

$$\sum_{\ell=1}^{L_p} \sum_{i,a} |y_i^{(\ell, a)}| \|A_{a,*}\|_1$$

by the smaller realized sum

$$\sum_{\ell=1}^{L_p} \sum_a \left(|y_{q_\ell}^{(\ell, a), \text{mc}}| + |y_{r_\ell}^{(\ell, a), \text{mc}}| \right) \|A_{a,*}\|_1.$$

Consequently the modified-coordinate radius is non-vacuous and tends to zero as the roundoff quantities tend to zero with all dimensions, exact inputs, and realized randomness fixed.

Corollary 20 (All-coordinate multiply-one and subtract-zero FHT writeback specializations). *Keep the notation and hypotheses of the all-coordinate add-zero FHT specialization. Let*

$$\rho \in \{\text{mul1}, \text{sub0}\}$$

be a label for the two additional all-coordinate storage routines, and define the rounded storage map

$$c_{\text{mul1}}(z) = \text{fl}(z \cdot 1), \quad c_{\text{sub0}}(z) = \text{fl}(z - 0).$$

Both routines use the same exact signed-Hadamard preconditioner

$$P_\omega = H_p \text{diag}(\omega),$$

the same exact Sylvester/Walsh pair schedule $\mathcal{P}_p = ((q_\ell, r_\ell))_{\ell=1}^{L_p}$, and the same exact Rademacher and exact uniform-row probability laws as before. Their computed initial diagonal sign columns are still stored by the rounded multiply-one sign copy:

$$x_i^{(0,a),\rho} = \omega_a \mathbf{1}_{i=a}, \quad \hat{x}_i^{(0,a),\rho} = \begin{cases} \text{fl}(\omega_a \cdot 1), & i = a, \\ 0, & i \neq a, \end{cases} \quad E_i^{(0,a),\rho} = u \mathbf{1}_{i=a}.$$

For $\ell = 1, \dots, L_p$, write $q = q_\ell$ and $r = r_\ell$. The exact reference vector $x^{(\ell,a),\rho}$ is obtained by the exact FHT butterfly. The computed pre-writeback vector is

$$y_q^{(\ell,a),\rho} = \text{fl}(\hat{x}_q^{(\ell-1,a),\rho} + \hat{x}_r^{(\ell-1,a),\rho}), \quad y_r^{(\ell,a),\rho} = \text{fl}(\hat{x}_q^{(\ell-1,a),\rho} - \hat{x}_r^{(\ell-1,a),\rho}),$$

and $y_i^{(\ell,a),\rho} = \hat{x}_i^{(\ell-1,a),\rho}$ for $i \notin \{q, r\}$. Before the explicit copy/writeback, define

$$F_i^{(\ell,a),\rho} = \begin{cases} u|\hat{x}_q^{(\ell-1,a),\rho} + \hat{x}_r^{(\ell-1,a),\rho}| + E_q^{(\ell-1,a),\rho} + E_r^{(\ell-1,a),\rho}, & i = q, \\ u|\hat{x}_q^{(\ell-1,a),\rho} - \hat{x}_r^{(\ell-1,a),\rho}| + E_q^{(\ell-1,a),\rho} + E_r^{(\ell-1,a),\rho}, & i = r, \\ E_i^{(\ell-1,a),\rho}, & i \notin \{q, r\}. \end{cases}$$

The all-coordinate ρ -writeback step stores every coordinate by

$$\hat{x}_i^{(\ell,a),\rho} = c_\rho(y_i^{(\ell,a),\rho}), \quad E_i^{(\ell,a),\rho} = F_i^{(\ell,a),\rho} + u|y_i^{(\ell,a),\rho}|.$$

Thus both $z \mapsto \text{fl}(z \cdot 1)$ and $z \mapsto \text{fl}(z - 0)$ are charged as computed non-probability writebacks, while the random signs and sampled rows remain exact.

With

$$c_p = \sqrt{1/m}, \quad \hat{c}_p = \text{fl}(\sqrt{1/m}), \quad \eta_p = c_p u,$$

define

$$\begin{aligned} \hat{P}_{\omega,ia}^\rho &= \text{fl}(\hat{c}_p \hat{x}_i^{(L_p,a),\rho}), \\ E_{\omega,ia}^{P,\rho} &= u|\hat{c}_p \hat{x}_i^{(L_p,a),\rho}| + |\hat{c}_p| E_i^{(L_p,a),\rho} + \eta_p(|\hat{x}_i^{(L_p,a),\rho}| + E_i^{(L_p,a),\rho}). \end{aligned}$$

Then

$$|\hat{P}_{\omega,ia}^\rho - (P_\omega)_{ia}| \leq E_{\omega,ia}^{P,\rho}.$$

Define

$$\hat{Y}_\omega^\rho = \text{fl}(\hat{P}_\omega^\rho A), \quad E_{\omega,ij}^{Y,\rho} = \gamma_m \sum_{a=1}^m |\hat{P}_{\omega,ia}^\rho| |A_{aj}| + \sum_{a=1}^m E_{\omega,ia}^{P,\rho} |A_{aj}|.$$

Using the same exact denominator $d = \sqrt{s/m}$, the same computed $\hat{d}_\star$, and the same

$$\delta_d = \frac{4u + 3u^2 + u^3}{1 - u},$$

define

$$\tilde{B}_{\omega,J,tj}^\rho = \frac{(\hat{Y}_\omega^\rho)_{J,tj}}{d}, \quad \hat{B}_{\omega,J,tj}^\rho = \mathfrak{fl}\left(\frac{(\hat{Y}_\omega^\rho)_{J,tj}}{\hat{d}_\star}\right),$$

$$\hat{G}_A^\rho(\omega, J) = \mathfrak{fl}\left((\hat{B}_{\omega,J}^\rho)^T \hat{B}_{\omega,J}^\rho\right),$$

and

$$E_{tj}^{\text{den},\rho} = |\hat{B}_{\omega,J,tj}^\rho| u + \frac{|(\hat{Y}_\omega^\rho)_{J,tj}| d \delta_d}{|\hat{d}_\star| d}.$$

The denominator, dot-product, and preconditioned-product contributions are

$$\tau_{\text{den},\rho}(\omega, J) = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den},\rho} |\hat{B}_{\omega,J,tj}^\rho| + |\tilde{B}_{\omega,J,tj}^\rho| E_{tk}^{\text{den},\rho} \right)_{j,k=1}^n \right\|_F,$$

$$\tau_{\text{dot},\rho}(\omega, J) = \left\| \left(\gamma_s \sum_{t=1}^s |\hat{B}_{\omega,J,tj}^\rho| |\hat{B}_{\omega,J,tk}^\rho| \right)_{j,k=1}^n \right\|_F,$$

and

$$\tau_{\text{prod},\rho}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega,J,tj}^{Y,\rho}}{d} |\tilde{B}_{\omega,J,tj}^\rho| + |B_{\omega,J,tj}| \frac{E_{\omega,J,tk}^{Y,\rho}}{d} \right)_{j,k=1}^n \right\|_F.$$

Set

$$\tau_{A,\text{syl},\rho}^{\text{cd}}(\omega, J) = \tau_{\text{den},\rho}(\omega, J) + \tau_{\text{dot},\rho}(\omega, J) + \tau_{\text{prod},\rho}(\omega, J).$$

Under the same exact preprocessing and sample-budget condition,

$$\mathbb{P}\left[(1 - \varepsilon)A^T A - \tau_{A,\text{syl},\rho}^{\text{cd}}(\omega, J)I_n \preceq \hat{G}_A^\rho(\omega, J) \preceq (1 + \varepsilon)A^T A + \tau_{A,\text{syl},\rho}^{\text{cd}}(\omega, J)I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

For fixed p, m, n, s, A, ω, J , the irreducible all-coordinate copy contribution is

$$\sum_{\ell=1}^{L_p} \sum_{i,a} |y_i^{(\ell,a),\rho}| \|A_{a,*}\|_1.$$

In the small-roundoff regime $\gamma_m = \Theta(mu)$, $\gamma_s = \Theta(su)$, and $\delta_d = \Theta(u)$,

$$\begin{aligned} \tau_{A,\text{syl},\rho}^{\text{cd}}(\omega, J) = O\bigg((u + \gamma_m + \gamma_s + \delta_d) \bigg[\sum_{i,a} |\hat{x}_i^{(L_p,a),\rho}| \|A_{a,*}\|_1 \\ + \sum_{\ell=1}^{L_p} \sum_{i,a} |y_i^{(\ell,a),\rho}| \|A_{a,*}\|_1 + \sum_{t,j,k} |\hat{B}_{\omega,J,tj}^\rho| |\hat{B}_{\omega,J,tk}^\rho| \bigg] \bigg). \end{aligned}$$

The same expression is a Θ -order description whenever the bracketed realized magnitude is bounded above and below by positive constants independent of u . Consequently both mul1 and sub0 radii are non-vacuous and tend to zero as the roundoff quantities tend to zero with all dimensions, exact inputs, and realized randomness fixed.

Corollary 21 (Modified-coordinate multiply-one and subtract-zero FHT writeback specializations). *Keep the notation and hypotheses of the modified-coordinate add-zero specialization and the all-coordinate mull/sub0 specialization. Let*

$$\rho \in \{\text{mull}, \text{sub0}\}, \quad c_{\text{mull}}(z) = \text{fl}(z \cdot 1), \quad c_{\text{sub0}}(z) = \text{fl}(z - 0).$$

The exact signed-Hadamard preconditioner is still

$$P_\omega = H_p \text{diag}(\omega),$$

and the pair schedule is still

$$\mathcal{P}_p = ((q_\ell, r_\ell))_{\ell=1}^{L_p}.$$

For each column a , initialize

$$x_i^{(0,a),\text{mc},\rho} = \omega_a \mathbf{1}_{i=a}, \quad \hat{x}_i^{(0,a),\text{mc},\rho} = \begin{cases} \text{fl}(\omega_a \cdot 1), & i = a, \\ 0, & i \neq a, \end{cases} \quad E_i^{(0,a),\text{mc},\rho} = u \mathbf{1}_{i=a}.$$

For the ℓ -th pair, write $q = q_\ell$ and $r = r_\ell$. The exact reference vector $x^{(\ell,a),\text{mc},\rho}$ is obtained by the exact FHT butterfly. The computed pre-writeback vector is

$$y_q^{(\ell,a),\text{mc},\rho} = \text{fl}(\hat{x}_q^{(\ell-1,a),\text{mc},\rho} + \hat{x}_r^{(\ell-1,a),\text{mc},\rho}),$$

$$y_r^{(\ell,a),\text{mc},\rho} = \text{fl}(\hat{x}_q^{(\ell-1,a),\text{mc},\rho} - \hat{x}_r^{(\ell-1,a),\text{mc},\rho}),$$

and $y_i^{(\ell,a),\text{mc},\rho} = \hat{x}_i^{(\ell-1,a),\text{mc},\rho}$ for $i \notin \{q, r\}$. Before the explicit modified-coordinate writeback, define

$$F_i^{(\ell,a),\text{mc},\rho} = \begin{cases} u|\hat{x}_q^{(\ell-1,a),\text{mc},\rho} + \hat{x}_r^{(\ell-1,a),\text{mc},\rho}| + E_q^{(\ell-1,a),\text{mc},\rho} + E_r^{(\ell-1,a),\text{mc},\rho}, & i = q, \\ u|\hat{x}_q^{(\ell-1,a),\text{mc},\rho} - \hat{x}_r^{(\ell-1,a),\text{mc},\rho}| + E_q^{(\ell-1,a),\text{mc},\rho} + E_r^{(\ell-1,a),\text{mc},\rho}, & i = r, \\ E_i^{(\ell-1,a),\text{mc},\rho}, & i \notin \{q, r\}. \end{cases}$$

The modified-coordinate ρ -writeback stores only the two butterfly outputs:

$$\hat{x}_i^{(\ell,a),\text{mc},\rho} = \begin{cases} c_\rho(y_i^{(\ell,a),\text{mc},\rho}), & i \in \{q, r\}, \\ y_i^{(\ell,a),\text{mc},\rho}, & i \notin \{q, r\}, \end{cases}$$

$$E_i^{(\ell,a),\text{mc},\rho} = \begin{cases} F_i^{(\ell,a),\text{mc},\rho} + u|y_i^{(\ell,a),\text{mc},\rho}|, & i \in \{q, r\}, \\ F_i^{(\ell,a),\text{mc},\rho}, & i \notin \{q, r\}. \end{cases}$$

Thus $z \mapsto \text{fl}(z \cdot 1)$ and $z \mapsto \text{fl}(z - 0)$ are charged only on modified coordinates; untouched coordinates carry their prior propagated error without a writeback term.

With

$$c_p = \sqrt{1/m}, \quad \hat{c}_p = \text{fl}(\sqrt{1/m}), \quad \eta_p = c_p u,$$

define

$$\hat{P}_{\omega,ia}^{\text{mc},\rho} = \text{fl}(\hat{c}_p \hat{x}_i^{(L_p,a),\text{mc},\rho}),$$

$$E_{\omega,ia}^{P,\text{mc},\rho} = u|\hat{c}_p \hat{x}_i^{(L_p,a),\text{mc},\rho}| + |\hat{c}_p| E_i^{(L_p,a),\text{mc},\rho} + \eta_p (|\hat{x}_i^{(L_p,a),\text{mc},\rho}| + E_i^{(L_p,a),\text{mc},\rho}).$$

Then

$$|\hat{P}_{\omega,ia}^{\text{mc},\rho} - (P_\omega)_{ia}| \leq E_{\omega,ia}^{P,\text{mc},\rho}.$$

Set

$$\begin{aligned}\widehat{Y}_\omega^{\text{mc},\rho} &= \text{fl}(\widehat{P}_\omega^{\text{mc},\rho} A), \\ E_{\omega,ij}^{Y,\text{mc},\rho} &= \gamma_m \sum_{a=1}^m |\widehat{P}_{\omega,ia}^{\text{mc},\rho}| |A_{aj}| + \sum_{a=1}^m E_{\omega,ia}^{P,\text{mc},\rho} |A_{aj}|.\end{aligned}$$

Using the same $d = \sqrt{s/m}$, $\widehat{d}_\star$, and $\delta_d = (4u + 3u^2 + u^3)/(1 - u)$, define

$$\widetilde{B}_{\omega,J,tj}^{\text{mc},\rho} = \frac{(\widehat{Y}_\omega^{\text{mc},\rho})_{J,tj}}{d}, \quad \widehat{B}_{\omega,J,tj}^{\text{mc},\rho} = \text{fl}\left(\frac{(\widehat{Y}_\omega^{\text{mc},\rho})_{J,tj}}{\widehat{d}_\star}\right),$$

$$\widehat{G}_A^{\text{mc},\rho}(\omega, J) = \text{fl}\left((\widehat{B}_{\omega,J}^{\text{mc},\rho})^T \widehat{B}_{\omega,J}^{\text{mc},\rho}\right).$$

Let

$$E_{tj}^{\text{den},\text{mc},\rho} = |\widehat{B}_{\omega,J,tj}^{\text{mc},\rho}| u + \frac{|(\widehat{Y}_\omega^{\text{mc},\rho})_{J,tj}| d \delta_d}{|\widehat{d}_\star| d}.$$

Define

$$\begin{aligned}\tau_{\text{den},\text{mc},\rho}(\omega, J) &= \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den},\text{mc},\rho} |\widehat{B}_{\omega,J,tk}^{\text{mc},\rho}| + |\widetilde{B}_{\omega,J,tj}^{\text{mc},\rho}| E_{tk}^{\text{den},\text{mc},\rho} \right)_{j,k=1}^n \right\|_F, \\ \tau_{\text{dot},\text{mc},\rho}(\omega, J) &= \left\| \left(\gamma_s \sum_{t=1}^s |\widehat{B}_{\omega,J,tj}^{\text{mc},\rho}| |\widehat{B}_{\omega,J,tk}^{\text{mc},\rho}| \right)_{j,k=1}^n \right\|_F,\end{aligned}$$

and

$$\tau_{\text{prod},\text{mc},\rho}(\omega, J) = \left\| \left(\sum_{t=1}^s \frac{E_{\omega,J,tj}^{Y,\text{mc},\rho}}{d} |\widetilde{B}_{\omega,J,tk}^{\text{mc},\rho}| + |B_{\omega,J,tj}| \frac{E_{\omega,J,tk}^{Y,\text{mc},\rho}}{d} \right)_{j,k=1}^n \right\|_F.$$

The full modified-coordinate ρ -radius is

$$\tau_{A,\text{syl},\text{mc},\rho}^{\text{cd}}(\omega, J) = \tau_{\text{den},\text{mc},\rho}(\omega, J) + \tau_{\text{dot},\text{mc},\rho}(\omega, J) + \tau_{\text{prod},\text{mc},\rho}(\omega, J).$$

Under the same exact preprocessing and sample-budget condition,

$$\mathbb{P}\left[(1 - \varepsilon)A^T A - \tau_{A,\text{syl},\text{mc},\rho}^{\text{cd}}(\omega, J)I_n \preceq \widehat{G}_A^{\text{mc},\rho}(\omega, J) \preceq (1 + \varepsilon)A^T A + \tau_{A,\text{syl},\text{mc},\rho}^{\text{cd}}(\omega, J)I_n\right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

For fixed p, m, n, s, A, ω, J , the modified-coordinate copy contribution is

$$\sum_{\ell=1}^{L_p} \sum_a \left(|y_{q_\ell}^{(\ell,a),\text{mc},\rho}| + |y_{r_\ell}^{(\ell,a),\text{mc},\rho}| \right) \|A_{a,*}\|_1.$$

In the small-roundoff regime $\gamma_m = \Theta(mu)$, $\gamma_s = \Theta(su)$, and $\delta_d = \Theta(u)$,

$$\begin{aligned}\tau_{A,\text{syl},\text{mc},\rho}^{\text{cd}}(\omega, J) &= O\left((u + \gamma_m + \gamma_s + \delta_d) \left[\sum_{i,a} |\widehat{x}_i^{(L_p,a),\text{mc},\rho}| \|A_{a,*}\|_1 \right. \right. \\ &\quad + \sum_{\ell=1}^{L_p} \sum_a \left(|y_{q_\ell}^{(\ell,a),\text{mc},\rho}| + |y_{r_\ell}^{(\ell,a),\text{mc},\rho}| \right) \|A_{a,*}\|_1 \\ &\quad \left. \left. + \sum_{t,j,k} |\widehat{B}_{\omega,J,tj}^{\text{mc},\rho}| |\widehat{B}_{\omega,J,tk}^{\text{mc},\rho}| \right] \right).\end{aligned}$$

The same expression is a Θ -order description whenever the bracketed realized magnitude is bounded above and below by positive constants independent of u . Consequently both modified-coordinate `mul1` and modified-coordinate `sub0` radii are non-vacuous and tend to zero as the roundoff quantities tend to zero with all dimensions, exact inputs, and realized randomness fixed.

For the modeled functional FHT schedules, the formalized same-stage no-alias/order facts show that distinct generated butterfly pairs commute for the rounded base update and for all three modified-coordinate writeback routines. These facts justify reordering distinct pairs inside one generated stage for this model; they do not assert arbitrary in-place, vectorized, or overwrite-array semantics. The same certificates are also specialized to the concrete Sylvester/Walsh stage list $\text{Pairs}_{\text{syl}}(p, r)$, so the 2^p -dimensional generated stage interface can be used directly without unfolding to a generic stride schedule. The matching no-touch facts for the rounded base schedule and the three modified-coordinate writeback schedules say that any coordinate outside every scheduled pair is returned unchanged. This is why the modified-coordinate copy contribution appears only on pair outputs; the statement is not claimed for all-coordinate writeback routines, which rewrite every coordinate by design. The corresponding propagated-budget certificates are also proved: on those same untouched coordinates, the base rounded budget and the three modified-coordinate budgets remain exactly equal to the incoming radius E_i . The same value and propagated-budget certificates are specialized to the concrete Sylvester/Walsh generated interfaces: the full generated schedule $\text{Pairs}_{\text{syl}}(p)$ and a single generated stage $\text{Pairs}_{\text{syl}}(p, r)$. Thus, whenever a coordinate is outside every pair in the relevant generated list, the full or one-stage rounded base schedule and the three modified-coordinate schedules return the incoming value, and their propagated budgets return exactly the incoming radius. This adds no probability-law term and no additional roundoff radius; it is a concrete interface wrapper for the already charged functional schedules.

Corollary 22 (Readable deterministic cap for computed SRHT on the Algorithm 3 input). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix handed to Algorithm 3. Let $0 < \delta_{\text{pre}} < m$ be the target SRHT preprocessing failure budget. Let $0 < \delta_{\text{sample}} < 1$ be the target uniform-row sampling failure budget. Fix an exact distortion tolerance $0 < \varepsilon < 1$, and set*

$$t_{\text{pre}} = \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}.$$

This is the standard logarithmic choice from Tropp's SRHT row-norm analysis [2]. It makes

$$m \exp\left(-\frac{mt_{\text{pre}}^2}{8}\right) = \delta_{\text{pre}}.$$

Assume first that $A \neq 0$. Let

$$r = \text{rank}(A)$$

and choose exact analysis-only matrices $U_A \in \mathbb{R}^{m \times r}$ and $C_A \in \mathbb{R}^{r \times n}$ such that

$$U_A^T U_A = I_r, \quad A = U_A C_A.$$

The matrix U_A is not computed by the algorithm in this theorem. It is only the exact column-space basis used to state the SRHT probability event. This corollary is the fixed-radius readable cap obtained by bounding the sample-dependent radius $\tau_A^{\text{cd}}(\omega, J)$ from the preceding theorem on the same exact SRHT row-norm event.

Let $\mathcal{S} \in \mathbb{R}^{m \times m}$ be an exact sign-pattern table with $\mathcal{S}_{ia}^2 = 1$, and let $\omega \in \{\pm 1\}^m$ be the exact realized Rademacher sign vector. Store the signs by one of the rounded copy routines $\text{fl}(\omega_a \cdot 1)$,

$\text{fl}(\omega_a + 0)$, or $\text{fl}(\omega_a - 0)$, and write the stored signs as $\hat{\omega}^\rho$, so that

$$|\hat{\omega}_a^\rho - \omega_a| \leq u.$$

Define

$$\begin{aligned} c_m &= \sqrt{1/m}, & \hat{c}_m &= \text{fl}(\sqrt{1/m}), \\ H_{ia} &= c_m S_{ia}, & \hat{H}_{ia} &= \hat{c}_m S_{ia}, \\ D_\omega &= \text{diag}(\omega), & \hat{D}_\omega^\rho &= \text{diag}(\hat{\omega}^\rho), \end{aligned}$$

and

$$P_\omega = H D_\omega, \quad \hat{P}_\omega^\rho = \text{fl}(\hat{H} \hat{D}_\omega^\rho).$$

Then P_ω is exact arithmetic and orthogonal, while $\hat{P}_\omega^\rho$ is the computed materialized preconditioner. The computed preconditioner is charged by the rounded-scale, stored-sign, and rounded $H \text{diag}(\omega)$ product budgets displayed in the preceding theorem.

Define the exact and computed Algorithm 3 preconditioned matrices

$$Y_\omega = P_\omega A, \quad \hat{Y}_\omega^\rho = \text{fl}(\hat{P}_\omega^\rho A).$$

For every row i and column j ,

$$|\hat{Y}_{\omega,ij}^\rho - (Y_\omega)_{ij}| \leq \kappa_m \|A_{*j}\|_1.$$

Thus the preconditioned matrix bound is for the actual input matrix A , not for an assumed orthonormal input.

Define the column-size parameters

$$a_1 = \max_{1 \leq j \leq n} \|A_{*j}\|_1, \quad a_2 = \max_{1 \leq j \leq n} \|A_{*j}\|_2, \quad \mu_A = \frac{a_1}{\sqrt{m} a_2}.$$

Since $A \neq 0$, $a_2 > 0$; also $0 < \mu_A \leq 1$. Define the rank-sensitive logarithmic SRHT radius

$$S_{A,\log}(\delta_{\text{pre}}) = \sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}, \quad L_{A,\log}(\delta_{\text{pre}}) = m S_{A,\log}(\delta_{\text{pre}})^2,$$

choose the exact row-sample count

$$s = s_{\min}(L_{A,\log}(\delta_{\text{pre}}), \varepsilon, \delta_{\text{sample}}; r),$$

and choose any exact Chernoff parameter $\theta_A > 0$ witnessing this definition, so that

$$\Phi(\theta_A, L_{A,\log}(\delta_{\text{pre}}), \varepsilon, s, r) \leq \delta_{\text{sample}}.$$

For a standard constant-distortion specialization, take $\varepsilon = 1/2$; then

$$s = s_{\min}(L_{A,\log}(\delta_{\text{pre}}), 1/2, \delta_{\text{sample}}; r),$$

and the literature-order sufficient scaling is

$$s = O\left(\left(\sqrt{r} + \sqrt{8 \log(m/\delta_{\text{pre}})}\right)^2 \log \frac{r}{\delta_{\text{sample}}}\right).$$

For symbolic $0 < \varepsilon < 1$, the same rule gives

$$s = O\left(\frac{(\sqrt{r} + \sqrt{8 \log(m/\delta_{\text{pre}})})^2}{\varepsilon^2} \log \frac{r}{\delta_{\text{sample}}}\right).$$

This is an exact parameter-selection step, not a floating-point computation. Define the explicit preconditioner-product roundoff scalar

$$K_m = \gamma_m(2 + \gamma_m)(1 + u)^2 + u(1 + u) + u.$$

Equivalently, $\kappa_m = K_m/\sqrt{m}$. Let

$$d = \sqrt{s/m}, \quad \hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{m}))),$$

and

$$\delta_d = \frac{4u + 3u^2 + u^3}{1 - u}, \quad 0 \leq \delta_d < 1.$$

The denominator certificate proves $\hat{d}_\star \neq 0$ and $|\hat{d}_\star - d| \leq d\delta_d$. On the exact SRHT row-norm event for U_A , every exact scaled input-sketch entry satisfies

$$\left| \frac{(Y_\omega)_{J_{tj}}}{d} \right| \leq \sqrt{\frac{m}{s}} S_{A, \log}(\delta_{\text{pre}}) a_2.$$

Here the column norm of C_A equals the column norm of A , because $U_A^T U_A = I_r$ and $A = U_A C_A$. Every corresponding computed scaled entry differs from it by at most the same quantity times

$$\zeta_{A, \log}(\delta_{\text{pre}}) = \frac{u + \delta_d + (1 + u)\mu_A K_m / S_{A, \log}(\delta_{\text{pre}})}{1 - \delta_d}.$$

Let

$$B_{\omega, J, tj} = \frac{(Y_\omega)_{J_{tj}}}{d}, \quad \hat{B}_{\omega, J, tj}^\rho = \text{fl}\left(\hat{Y}_{\omega, J_{tj}}^\rho / \hat{d}_\star\right).$$

The exact sampled input Gram and computed sampled input Gram are

$$G_A(\omega, J) = B_{\omega, J}^T B_{\omega, J}, \quad \hat{G}_A^\rho(\omega, J) = \text{fl}\left((\hat{B}_{\omega, J}^\rho)^T \hat{B}_{\omega, J}^\rho\right).$$

Conditioned on ω , the exact row sample is unbiased:

$$\mathbb{E}_J[G_A(\omega, J) \mid \omega] = (P_\omega A)^T (P_\omega A) = A^T A.$$

This identity is exact arithmetic; it is not computed by the algorithm.

Define the final floating-point radius by the fully expanded formula

$$\begin{aligned}
T_{A,\log}^{\text{fp}}(\delta_{\text{pre}}) &= nm \left(\sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right)^2 \\
&\times \left(\max_{1 \leq j \leq n} \|A_{*j}\|_2 \right)^2 \\
&\times \left[2 \frac{u + \frac{4u+3u^2+u^3}{1-u} + (1+u) \frac{\max_{1 \leq j \leq n} \|A_{*j}\|_1}{\sqrt{m} \max_{1 \leq j \leq n} \|A_{*j}\|_2} \frac{\gamma_m(2+\gamma_m)(1+u)^2+u(1+u)+u}{\sqrt{r/m} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}}}{1 - \frac{4u+3u^2+u^3}{1-u}} \right. \\
&+ \left. \left(\frac{u + \frac{4u+3u^2+u^3}{1-u} + (1+u) \frac{\max_{1 \leq j \leq n} \|A_{*j}\|_1}{\sqrt{m} \max_{1 \leq j \leq n} \|A_{*j}\|_2} \frac{\gamma_m(2+\gamma_m)(1+u)^2+u(1+u)+u}{\sqrt{r/m} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}}}{1 - \frac{4u+3u^2+u^3}{1-u}} \right)^2 \right. \\
&+ \left. \gamma_s \left(1 + \frac{u + \frac{4u+3u^2+u^3}{1-u} + (1+u) \frac{\max_{1 \leq j \leq n} \|A_{*j}\|_1}{\sqrt{m} \max_{1 \leq j \leq n} \|A_{*j}\|_2} \frac{\gamma_m(2+\gamma_m)(1+u)^2+u(1+u)+u}{\sqrt{r/m} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}}}}{1 - \frac{4u+3u^2+u^3}{1-u}} \right)^2 \right].
\end{aligned}$$

This is exactly the compact expression

$$nm S_{A,\log}(\delta_{\text{pre}})^2 a_2^2 (2\zeta_{A,\log}(\delta_{\text{pre}}) + \zeta_{A,\log}(\delta_{\text{pre}})^2 + \gamma_s(1 + \zeta_{A,\log}(\delta_{\text{pre}})))^2,$$

but all locally introduced shorthand has been recursively replaced in the displayed definition of $T_{A,\log}^{\text{fp}}(\delta_{\text{pre}})$, including the concrete denominator relative error $(4u + 3u^2 + u^3)/(1 - u)$. The exact-denominator variant is available as infrastructure by replacing this relative denominator term by 0; it is not the selected final concrete endpoint in this corollary. On the same exact SRHT row-norm event,

$$\left\| \widehat{G}_A^\rho(\omega, J) - G_A(\omega, J) \right\|_F \leq T_{A,\log}^{\text{fp}}(\delta_{\text{pre}}).$$

By construction of the exact sample count s and witness θ_A ,

$$\Phi(\theta_A, L_{A,\log}(\delta_{\text{pre}}), \varepsilon, s, r) \leq \delta_{\text{sample}}.$$

Then the fully computed Algorithm 3 sampled input Gram satisfies

$$\mathbb{P} \left[(1 - \varepsilon) A^T A - T_{A,\log}^{\text{fp}}(\delta_{\text{pre}}) I_n \preceq \widehat{G}_A^\rho(\omega, J) \preceq (1 + \varepsilon) A^T A + T_{A,\log}^{\text{fp}}(\delta_{\text{pre}}) I_n \right] \geq 1 - \delta_{\text{pre}} - \delta_{\text{sample}}.$$

The probability uses only the exact Rademacher and exact uniform-row laws. No probability table is computed. The non-probability computations charged in $T_{A,\log}^{\text{fp}}(\delta_{\text{pre}})$ are the rounded scale $\widehat{c}_m$, rounded sign storage, rounded formation of $\widehat{P}_\omega^\rho$, rounded formation of $\widehat{Y}_\omega^\rho = \text{fl}(\widehat{P}_\omega^\rho A)$, rounded denominator $\widehat{d}_*$, rounded row scaling, and rounded Gram dot products.

In the small-roundoff regime

$$0 \leq u, \gamma_m, \gamma_s, \frac{4u + 3u^2 + u^3}{1 - u} \leq c < 1, \quad \sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} > 0, \quad A \neq 0,$$

with $m, n, r, s, \delta_{\text{pre}}$ and the column norms of A fixed, the expanded relative error fraction in $T_{A, \log}^{\text{fp}}(\delta_{\text{pre}})$ has order

$$\Theta \left(u + \frac{4u + 3u^2 + u^3}{1 - u} + \frac{(\gamma_m + u) \max_{1 \leq j \leq n} \|A_{*j}\|_1}{\sqrt{m} \left(\sqrt{r/m} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right) \max_{1 \leq j \leq n} \|A_{*j}\|_2} \right),$$

and

$$T_{A, \log}^{\text{fp}}(\delta_{\text{pre}}) = \Theta \left(nm \left(\sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right)^2 \right. \\ \left. \times \left(\max_{1 \leq j \leq n} \|A_{*j}\|_2 \right)^2 \left[\gamma_s + u + \frac{4u + 3u^2 + u^3}{1 - u} + \frac{(\gamma_m + u) \max_{1 \leq j \leq n} \|A_{*j}\|_1}{\sqrt{m} \left(\sqrt{r/m} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right) \max_{1 \leq j \leq n} \|A_{*j}\|_2} \right] \right).$$

Equivalently,

$$T_{A, \log}^{\text{fp}}(\delta_{\text{pre}}) = \Theta \left(nm \left(\sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right)^2 \right. \\ \times \left(\max_{1 \leq j \leq n} \|A_{*j}\|_2 \right)^2 \left(\gamma_s + u + \frac{4u + 3u^2 + u^3}{1 - u} \right) \\ \left. + n\sqrt{m} \left(\sqrt{\frac{r}{m}} + \sqrt{\frac{8 \log(m/\delta_{\text{pre}})}{m}} \right) \left(\max_{1 \leq j \leq n} \|A_{*j}\|_1 \right) \left(\max_{1 \leq j \leq n} \|A_{*j}\|_2 \right) (\gamma_m + u) \right).$$

For fixed $A, m, n, r, s, \delta_{\text{pre}}$,

$$T_{A, \log}^{\text{fp}}(\delta_{\text{pre}}) \longrightarrow 0 \quad \text{as} \quad u, \gamma_m, \gamma_s, \frac{4u + 3u^2 + u^3}{1 - u} \longrightarrow 0,$$

so the result is non-vacuous. If $A = 0$, then $a_1 = a_2 = 0$, $A^T A = 0$, the exact and computed preconditioned products are zero in the floating-point model, and the theorem holds with $T_{A, \log}^{\text{fp}}(\delta_{\text{pre}}) = 0$.

CountSketch Sparse-Gram Endpoints

This section records the currently closed CountSketch branches for Algorithm 3. The CountSketch transform is the usual sparse embedding pattern [3, 4]. The first closed route below is collision-free, with probability loss m^2/r ; it is non-vacuous only when r is larger than m^2 . The later product-grid route is the local finite-cover Markov/Frobenius route. Neither route should be read as the optimal Clarkson–Woodruff subspace-embedding concentration theorem.

Let r be the number of CountSketch buckets. A hash

$$h : \{1, \dots, m\} \rightarrow \{1, \dots, r\}$$

and a sign vector $\omega \in \{\pm 1\}^m$ are sampled from the exact CountSketch product law. Sampling probabilities are exact mathematical laws in this note. Define the exact sparse CountSketch matrix

$$S_{h, \omega}(i, k) = \begin{cases} \omega_k, & h(k) = i, \\ 0, & h(k) \neq i, \end{cases}$$

and the exact sketched matrix

$$Y(h, \omega; A) = S_{h, \omega} A.$$

The exact CountSketch Gram error is

$$\Delta_{\text{CS}}(h, \omega; A) = Y(h, \omega; A)^T Y(h, \omega; A) - A^T A.$$

The computed sparse matrix $\hat{Y}$ is formed entrywise by multiplying each selected input entry by its exact sign in floating point and then accumulating each bucket in floating point. Let

$$b_i(h) = \#\{k : h(k) = i\}$$

be the exact bucket size. The entrywise sparse-apply error radius is

$$E_{ij}^{\text{CS}}(h, \omega; A) = (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{k: h(k)=i} |\omega_k| |A_{kj}|.$$

This is computable from the exact realized hash/sign pattern and the exact input magnitudes. It charges the rounded signed products and the rounded bucket accumulation. Let $|Y|$ denote entrywise absolute value. The rounded sparse-Gram matrix is

$$\hat{G}(h, \omega; A) = \text{fl}(\hat{Y}^T \hat{Y}),$$

where each Gram entry is a rounded length- r dot product. The fully expanded sparse-Gram floating-point radius is

$$\begin{aligned} T_{\text{CSGram}}^{\text{fp}}(h, \omega; A) := & \left\| \left(\gamma_r \sum_{i=1}^r (|Y_{ij}| + E_{ij}^{\text{CS}}) (|Y_{ik}| + E_{ik}^{\text{CS}}) \right)_{j,k=1}^n \right\|_F^n \\ & + \left\| \left(\sum_{i=1}^r [E_{ij}^{\text{CS}} |Y_{ik}| + |Y_{ij}| E_{ik}^{\text{CS}} + E_{ij}^{\text{CS}} E_{ik}^{\text{CS}}] \right)_{j,k=1}^n \right\|_F^n. \end{aligned}$$

For fixed h, ω, A , valid γ_r , and small bucket roundoff,

$$T_{\text{CSGram}}^{\text{fp}} = \Theta(\gamma_r \| |Y|^T |Y| \|_F + \|(E^{\text{CS}})^T |Y| + |Y|^T E^{\text{CS}}\|_F)$$

when the displayed leading term is nonzero. Since $E_{ij}^{\text{CS}} = \Theta((u + \gamma_{b_i}) \sum_{k: h(k)=i} |A_{kj}|)$, the radius vanishes as $u, \gamma_r, \gamma_{b_i} \rightarrow 0$ for fixed realized hash/sign data and fixed A .

Collision-Free Actual-Input Route

Let q be an exact analysis dimension. Let

$$U \in \mathbb{R}^{m \times q}, \quad C \in \mathbb{R}^{q \times n}, \quad U^T U = I_q, \quad A = UC.$$

The matrices U and C are exact analysis witnesses for the column space of the input. Algorithm 3 is not assumed to compute them in this theorem; the implemented arithmetic below uses the actual input entries of A .

The collision-free event is

$$h(a) \neq h(b) \quad \text{for every } a \neq b.$$

Under the exact CountSketch hash law,

$$\mathbb{P}_h[h \text{ is collision-free}] \geq 1 - \frac{m^2}{r}.$$

On this event, the exact sparse transform preserves all Gram matrices:

$$Y(h, \omega; A)^T Y(h, \omega; A) = A^T A.$$

The Rademacher signs remain exact mathematical signs; only their use in floating-point multiplications is charged.

After the exact CountSketch draw, let

$$\sigma_1, \dots, \sigma_s \in \{1, \dots, r\}$$

be exact iid uniform row samples. Let $\delta_{\text{samp}} \geq 0$ be such that

$$\Phi(\theta, r, \varepsilon, s, q) \leq \delta_{\text{samp}}, \quad \theta > 0.$$

Here the uniform-row Loewner radius parameter is $L = r$, because a collision-free CountSketch applied to an orthonormal U has orthonormal columns and every row has squared norm at most one. Thus a uniform row sample from the r -row preconditioned matrix has one-sample Loewner radius r .

Set

$$d = \sqrt{s/r}, \quad \hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{r}))) ,$$

and

$$\Delta_\star = \frac{4u + 3u^2 + u^3}{1 - u}.$$

The concrete denominator routine proves

$$\hat{d}_\star \neq 0, \quad |\hat{d}_\star - d| \leq d \Delta_\star.$$

Define the exact and computed sampled rows

$$B_{tj} = \frac{Y_{\sigma_{tj}}}{d}, \quad \hat{B}_{tj} = \text{fl}\left(\frac{\hat{Y}_{\sigma_{tj}}}{\hat{d}_\star}\right), \quad \tilde{B}_{tj} = \frac{\hat{Y}_{\sigma_{tj}}}{d}.$$

The denominator-and-division entry radius is

$$E_{tj}^{\text{den}} = u|\hat{B}_{tj}| + \frac{|\hat{Y}_{\sigma_{tj}}|}{|\hat{d}_\star|} \Delta_\star.$$

The sparse-apply entry radius propagated to the sampled rows is

$$F_{tj} = \frac{E_{\sigma_{tj}}^{\text{CS}}(h, \omega; A)}{d}.$$

The fully expanded downstream floating-point radius is

$$T_{\text{CF}, \text{row}}^{\text{fp}} = T_{\text{den}}^{\text{fp}} + T_{\text{dot}}^{\text{fp}} + T_{\text{basis}}^{\text{fp}},$$

where

$$T_{\text{den}}^{\text{fp}} = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den}} |\widehat{B}_{tk}| + |\widetilde{B}_{tj}| E_{tk}^{\text{den}} \right)_{j,k=1}^n \right\|_F,$$

$$T_{\text{dot}}^{\text{fp}} = \left\| \left(\gamma_s \sum_{t=1}^s |\widehat{B}_{tj}| |\widehat{B}_{tk}| \right)_{j,k=1}^n \right\|_F,$$

and

$$T_{\text{basis}}^{\text{fp}} = \left\| \left(\sum_{t=1}^s F_{tj} |\widetilde{B}_{tk}| + |B_{tj}| F_{tk} \right)_{j,k=1}^n \right\|_F.$$

Corollary 23 (Collision-free CountSketch actual-input sampled Gram). *Assume $r > 0$, $s > 0$, $U^T U = I_q$, $A = UC$, valid γ_m for the sparse bucket accumulations, valid γ_s for the sampled-Gram dot products, $0 < u < 1$, $\theta > 0$, and*

$$\Phi(\theta, r, \varepsilon, s, q) \leq \delta_{\text{samp}}, \quad \frac{m^2}{r} + \delta_{\text{samp}} \leq \delta.$$

Let

$$\widehat{G}_{\text{CF}}^{\text{fp}} = \text{fl}(\widehat{B}^T \widehat{B})$$

be the matrix produced by the implemented path: rounded sparse CountSketch apply to the actual input A , the concrete denominator $\widehat{d}_\star$, rounded sampled-row divisions, and rounded length- s Gram dot products. Then

$$\mathbb{P}_{h,\omega,\sigma} \left[-\varepsilon A^T A - T_{\text{CF,row}}^{\text{fp}} I_n \preceq \widehat{G}_{\text{CF}}^{\text{fp}} - A^T A \preceq \varepsilon A^T A + T_{\text{CF,row}}^{\text{fp}} I_n \right] \geq 1 - \delta.$$

The exact objects are A , the analysis witnesses U, C , the hash/sign law, the uniform row-sampling law, $Y = S_{h,\omega} A$, and $A^T A$. The computed non-probability objects are $\widehat{Y}$, $\widehat{d}_\star$, $\widehat{B}$, and $\widehat{G}_{\text{CF}}^{\text{fp}}$. There is no perturbation-event hypothesis, extra existence hypothesis, generic denominator parameter, or probability-computation error term.

For fixed realized h, ω, σ, A , fixed $d > 0$, and nonzero leading deterministic coefficient,

$$T_{\text{CF,row}}^{\text{fp}} = \Theta \left(\gamma_s \left\| |\widehat{B}|^T |\widehat{B}| \right\|_F + \left\| (E^{\text{den}})^T |\widehat{B}| + |\widetilde{B}|^T E^{\text{den}} \right\|_F \right. \\ \left. + \left\| F^T |\widetilde{B}| + |B|^T F \right\|_F \right),$$

with the irreducible substitutions

$$E_{tj}^{\text{den}} = \Theta \left(u |\widehat{B}_{tj}| + \frac{|\widehat{Y}_{\sigma_t j}|}{|\widehat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u} \right)$$

and

$$F_{tj} = \Theta \left(\frac{u + \gamma_{b_{\sigma_t}(h)}}{\sqrt{s/r}} \sum_{k:h(k)=\sigma_t} |A_{kj}| \right).$$

Consequently $T_{\text{CF,row}}^{\text{fp}} \rightarrow 0$ as

$$u, \gamma_s, \gamma_{b_i(h)}, \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0$$

for fixed realized data. The exact probability loss has order

$$\Theta\left(\frac{m^2}{r} + \Phi(\theta, r, \varepsilon, s, q)\right),$$

and is non-vacuous when $m^2/r + \Phi(\theta, r, \varepsilon, s, q) < 1$.

Stored-Sign Collision-Free Actual-Input Route

The preceding actual-input theorem uses exact Rademacher signs directly in the modeled sparse apply. If the implementation first stores the realized signs, the probability law is unchanged, but the stored signs are computed non-probability objects and their rounding must be charged.

Let the storage mode be

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

where

$$\hat{\omega}_k^{\rho_{\text{sig}}} \in \{\text{fl}(\omega_k \cdot 1), \text{fl}(\omega_k + 0), \text{fl}(\omega_k - 0)\}.$$

Since $|\omega_k| = 1$, each concrete mode satisfies

$$|\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| \leq u.$$

The exact objects remain $U, C, A = UC$, $Y = S_{h,\omega}A$, the exact hash/sign law, the exact uniform row-sampling law, and $A^T A$. The computed non-probability objects are now the stored sign table $\hat{\omega}^{\rho_{\text{sig}}}$, the rounded sparse apply $\hat{Y}^{\rho_{\text{sig}}}$, the concrete denominator $\hat{d}_\star$, rounded sampled-row divisions, and rounded sampled-Gram dot products.

For each CountSketch bucket i , define the stored-sign sparse-apply radius

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}}(h, \omega; A) &:= (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}}| |A_{kj}| \\ &+ \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| |A_{kj}|. \end{aligned}$$

After exact row samples $\sigma_1, \dots, \sigma_s$, set

$$B_{tj} = \frac{Y_{\sigma_t j}}{d}, \quad \hat{B}_{tj}^{\rho_{\text{sig}}} = \text{fl}\left(\frac{\hat{Y}_{\sigma_t j}^{\rho_{\text{sig}}}}{\hat{d}_\star}\right), \quad \tilde{B}_{tj}^{\rho_{\text{sig}}} = \frac{\hat{Y}_{\sigma_t j}^{\rho_{\text{sig}}}}{d}.$$

The denominator-and-division radius becomes

$$E_{tj}^{\text{den}, \rho_{\text{sig}}} = u|\hat{B}_{tj}^{\rho_{\text{sig}}}| + \frac{|\hat{Y}_{\sigma_t j}^{\rho_{\text{sig}}}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u},$$

and the sampled sparse-apply radius is

$$F_{tj}^{\rho_{\text{sig}}} = \frac{E_{\sigma_t j}^{\text{CS}, \rho_{\text{sig}}}(h, \omega; A)}{d}.$$

Define the stored-sign collision-free downstream radius by the irreducible sum

$$T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}}^{\text{fp}},$$

where

$$T_{\text{den}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\sum_{t=1}^s E_{tj}^{\text{den}, \rho_{\text{sig}}} |\widehat{B}_{tk}^{\rho_{\text{sig}}}| + |\widetilde{B}_{tj}^{\rho_{\text{sig}}}| E_{tk}^{\text{den}, \rho_{\text{sig}}} \right)_{j,k=1}^n \right\|_F,$$

$$T_{\text{dot}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\gamma_s \sum_{t=1}^s |\widehat{B}_{tj}^{\rho_{\text{sig}}}| |\widehat{B}_{tk}^{\rho_{\text{sig}}}| \right)_{j,k=1}^n \right\|_F,$$

and

$$T_{\text{basis}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\sum_{t=1}^s F_{tj}^{\rho_{\text{sig}}} |\widetilde{B}_{tk}^{\rho_{\text{sig}}}| + |B_{tj}| F_{tk}^{\rho_{\text{sig}}} \right)_{j,k=1}^n \right\|_F.$$

Corollary 24 (Stored-sign collision-free CountSketch actual-input sampled Gram). *Assume the hypotheses of the preceding collision-free actual-input corollary, and store every realized Rademacher sign by one of mul1, add0, sub0 before the sparse apply. Let*

$$\widehat{G}_{\text{CF}, \rho_{\text{sig}}}^{\text{fp}} = \text{fl} \left((\widehat{B}^{\rho_{\text{sig}}})^T \widehat{B}^{\rho_{\text{sig}}} \right)$$

be the matrix produced by the implemented path: rounded sign storage, rounded sparse CountSketch apply to the actual input A , the concrete denominator $\widehat{d}_\star$, rounded sampled-row divisions, and rounded length- s Gram dot products. Then

$$\mathbb{P}_{h, \omega, \sigma} \left[-\varepsilon A^T A - T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} I_n \preceq \widehat{G}_{\text{CF}, \rho_{\text{sig}}}^{\text{fp}} - A^T A \preceq \varepsilon A^T A + T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} I_n \right] \geq 1 - \delta.$$

No extra probability loss is introduced by sign storage; the storage errors enter only through the deterministic radius $T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}}$.

For fixed realized h, ω, σ, A , fixed $d > 0$, and nonzero leading deterministic coefficient,

$$T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} = \Theta \left(\gamma_s \left\| |\widehat{B}^{\rho_{\text{sig}}}|^T |\widehat{B}^{\rho_{\text{sig}}}| \right\|_F + \left\| (E^{\text{den}, \rho_{\text{sig}}})^T |\widehat{B}^{\rho_{\text{sig}}}| + |\widetilde{B}^{\rho_{\text{sig}}}|^T E^{\text{den}, \rho_{\text{sig}}} \right\|_F \right. \\ \left. + \left\| (F^{\rho_{\text{sig}}})^T |\widetilde{B}^{\rho_{\text{sig}}}| + |B|^T F^{\rho_{\text{sig}}} \right\|_F \right),$$

with the irreducible substitutions

$$E_{tj}^{\text{den}, \rho_{\text{sig}}} = \Theta \left(u |\widehat{B}_{tj}^{\rho_{\text{sig}}}| + \frac{|\widehat{Y}_{\sigma_t j}^{\rho_{\text{sig}}}|}{|\widehat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u} \right)$$

and

$$F_{tj}^{\rho_{\text{sig}}} = \Theta \left(\frac{u + \gamma_{b_{\sigma_t}(h)}}{\sqrt{s/r}} \sum_{k: h(k) = \sigma_t} |A_{kj}| \right).$$

Hence $T_{\text{CF}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} \rightarrow 0$ as

$$u, \gamma_s, \gamma_{b_i}(h), \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0$$

for fixed realized data. The probability loss is still

$$\Theta \left(\frac{m^2}{r} + \Phi(\theta, r, \varepsilon, s, q) \right),$$

so the result is non-vacuous under the same condition $m^2/r + \Phi(\theta, r, \varepsilon, s, q) < 1$.

Permuted-Bucket Stored-Sign Actual-Input Route

Some sparse CountSketch implementations traverse the entries in each realized bucket in a fixed order chosen by the implementation. For a realized hash h , let

$$\pi_i(h) : \{0, \dots, b_i(h) - 1\} \rightarrow \{0, \dots, b_i(h) - 1\}$$

be the exact permutation used to traverse bucket i , and let

$$k_{i,t}^\pi(h) := \text{bucketIndex}(h, i, \pi_i(h)(t))$$

be the original input-row index reached at traversal position t . The permutations $\pi_i(h)$ are exact discrete implementation choices. They are not random laws and are not floating-point real quantities, but the sparse signed products and bucket accumulations executed in that order are computed and must be charged.

With the same stored sign modes $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$, define the permuted sparse-apply entry radius

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}, \pi}(h, \omega; A) &:= (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{t=0}^{b_i(h)-1} |\hat{\omega}_{k_{i,t}^\pi(h)}^{\rho_{\text{sig}}}| |A_{k_{i,t}^\pi(h), j}| \\ &\quad + \sum_{t=0}^{b_i(h)-1} |\hat{\omega}_{k_{i,t}^\pi(h)}^{\rho_{\text{sig}}} - \omega_{k_{i,t}^\pi(h)}| |A_{k_{i,t}^\pi(h), j}|. \end{aligned}$$

Let $\hat{Y}^{\rho_{\text{sig}}, \pi}$ be the rounded sparse CountSketch apply to the actual input A , using the stored signs and the traversal orders $\pi_i(h)$. Define

$$\begin{aligned} \hat{B}_{tj}^{\rho_{\text{sig}}, \pi} &= \text{fl} \left(\frac{\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \pi}}{\hat{d}_\star} \right), \quad \tilde{B}_{tj}^{\rho_{\text{sig}}, \pi} = \frac{\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \pi}}{d}, \\ E_{tj}^{\text{den}, \rho_{\text{sig}}, \pi} &= u |\hat{B}_{tj}^{\rho_{\text{sig}}, \pi}| + \frac{|\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \pi}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u}, \end{aligned}$$

and

$$F_{tj}^{\rho_{\text{sig}}, \pi} = \frac{E_{\sigma_{tj}}^{\text{CS}, \rho_{\text{sig}}, \pi}(h, \omega; A)}{d}.$$

The irreducible permuted-bucket stored-sign downstream radius is

$$T_{\text{CF}, \text{row}, \rho_{\text{sig}}, \pi}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}, \pi}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}, \pi}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}, \pi}^{\text{fp}},$$

where

$$\begin{aligned} T_{\text{den}, \rho_{\text{sig}}, \pi}^{\text{fp}} &:= \left\| (E^{\text{den}, \rho_{\text{sig}}, \pi})^T |\hat{B}^{\rho_{\text{sig}}, \pi}| + |\tilde{B}^{\rho_{\text{sig}}, \pi}|^T E^{\text{den}, \rho_{\text{sig}}, \pi} \right\|_F, \\ T_{\text{dot}, \rho_{\text{sig}}, \pi}^{\text{fp}} &:= \gamma_s \left\| |\hat{B}^{\rho_{\text{sig}}, \pi}|^T |\hat{B}^{\rho_{\text{sig}}, \pi}| \right\|_F, \\ T_{\text{basis}, \rho_{\text{sig}}, \pi}^{\text{fp}} &:= \left\| (F^{\rho_{\text{sig}}, \pi})^T |\tilde{B}^{\rho_{\text{sig}}, \pi}| + |B|^T F^{\rho_{\text{sig}}, \pi} \right\|_F. \end{aligned}$$

Corollary 25 (Permuted-bucket stored-sign collision-free actual-input sampled Gram). *Assume the hypotheses of the stored-sign collision-free actual-input corollary, and fix the exact bucket traversal orders $\pi_i(h)$. Let*

$$\hat{G}_{\text{CF}, \rho_{\text{sig}}, \pi}^{\text{fp}} = \text{fl} \left((\hat{B}^{\rho_{\text{sig}}, \pi})^T \hat{B}^{\rho_{\text{sig}}, \pi} \right)$$

be the matrix produced by rounded sign storage, permuted-order rounded sparse CountSketch apply to the actual input A , the concrete denominator $\hat{d}_\star$, rounded sampled-row divisions, and rounded length- s Gram dot products. Then

$$\mathbb{P}_{h,\omega,\sigma} \left[-\varepsilon A^T A - T_{\text{CF,row},\rho_{\text{sig}},\pi}^{\text{fp}} I_n \preceq \hat{G}_{\text{CF},\rho_{\text{sig}},\pi}^{\text{fp}} - A^T A \preceq \varepsilon A^T A + T_{\text{CF,row},\rho_{\text{sig}},\pi}^{\text{fp}} I_n \right] \geq 1 - \delta.$$

Exact objects are $U, C, A = UC$, the hash/sign/row-sampling laws, the bucket orders $\pi_i(h)$, $Y = S_{h,\omega}A$, and $A^T A$. Computed non-probability objects are the stored signs, sparse signed products, permuted-order bucket accumulations, $\hat{d}_\star$, row divisions, and Gram dot products. No extra probability loss is introduced by the traversal order.

For fixed realized $h, \omega, \sigma, A, \pi$, fixed $d > 0$, and nonzero leading deterministic coefficient,

$$T_{\text{CF,row},\rho_{\text{sig}},\pi}^{\text{fp}} = \Theta \left(\gamma_s \left\| |\hat{B}^{\rho_{\text{sig}},\pi}|^T |\hat{B}^{\rho_{\text{sig}},\pi}| \right\|_F + \left\| (E^{\text{den},\rho_{\text{sig}},\pi})^T |\hat{B}^{\rho_{\text{sig}},\pi}| + |\tilde{B}^{\rho_{\text{sig}},\pi}|^T E^{\text{den},\rho_{\text{sig}},\pi} \right\|_F \right. \\ \left. + \left\| (F^{\rho_{\text{sig}},\pi})^T |\tilde{B}^{\rho_{\text{sig}},\pi}| + |B|^T F^{\rho_{\text{sig}},\pi} \right\|_F \right),$$

with the irreducible substitutions

$$E_{tj}^{\text{den},\rho_{\text{sig}},\pi} = \Theta \left(u |\hat{B}_{tj}^{\rho_{\text{sig}},\pi}| + \frac{|\hat{Y}_{\sigma_t j}^{\rho_{\text{sig}},\pi}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u} \right)$$

and

$$F_{tj}^{\rho_{\text{sig}},\pi} = \Theta \left(\frac{u + \gamma_{b_{\sigma_t}(h)}}{\sqrt{s/r}} \sum_{v=0}^{b_{\sigma_t}(h)-1} |A_{k_{\sigma_t,v}(h),j}| \right).$$

Therefore $T_{\text{CF,row},\rho_{\text{sig}},\pi}^{\text{fp}} \rightarrow 0$ as

$$u, \gamma_s, \gamma_{b_i(h)}, \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0$$

for fixed realized data. The probability loss is unchanged,

$$\Theta \left(\frac{m^2}{r} + \Phi(\theta, r, \varepsilon, s, q) \right),$$

so the result is non-vacuous under $m^2/r + \Phi(\theta, r, \varepsilon, s, q) < 1$.

Tree-Reduced Stored-Sign Actual-Input Route

Some sparse CountSketch implementations reduce each bucket by a fixed binary summation tree instead of a flat left-to-right accumulation. The tree shapes are exact discrete implementation choices. They are not random laws and are not floating-point real quantities, but the arithmetic performed along each tree is computed and must be charged.

For a realized hash h , let $\mathcal{T}_i(h)$ be the exact tree used for bucket i , and let $d_i^{\mathcal{T}}(h)$ be its depth. Assume the roundoff guard for every realized tree depth is valid, so $\gamma_{d_i^{\mathcal{T}}(h)}$ is defined. With the same stored sign modes $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$, define

$$E_{ij}^{\text{CS},\rho_{\text{sig}},\mathcal{T}}(h, \omega; A) := (u + \gamma_{d_i^{\mathcal{T}}(h)} + u \gamma_{d_i^{\mathcal{T}}(h)}) \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}}| |A_{kj}| \\ + \sum_{k:h(k)=i} |\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| |A_{kj}|.$$

This is the only change from the flat stored-sign radius: the bucket accumulation factor $\gamma_{b_i(h)}$ is replaced by the tree-depth factor $\gamma_{d_i^T(h)}$. Let $\hat{Y}^{\rho_{\text{sig}}, \mathcal{T}}$ be the rounded sparse CountSketch apply to the actual input A using the stored signs and the tree reductions. Define

$$\begin{aligned}\hat{B}_{tj}^{\rho_{\text{sig}}, \mathcal{T}} &= \text{fl}\left(\frac{\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \mathcal{T}}}{\hat{d}_\star}\right), & \tilde{B}_{tj}^{\rho_{\text{sig}}, \mathcal{T}} &= \frac{\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \mathcal{T}}}{d}, \\ E_{tj}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} &= u|\hat{B}_{tj}^{\rho_{\text{sig}}, \mathcal{T}}| + \frac{|\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \mathcal{T}}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u},\end{aligned}$$

and

$$F_{tj}^{\rho_{\text{sig}}, \mathcal{T}} = \frac{E_{\sigma_{tj}}^{\text{CS}, \rho_{\text{sig}}, \mathcal{T}}(h, \omega; A)}{d}.$$

The irreducible tree-reduced stored-sign downstream radius is

$$T_{\text{CF}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}},$$

where the three terms are obtained from the preceding stored-sign definitions by replacing $\hat{B}^{\rho_{\text{sig}}}, \tilde{B}^{\rho_{\text{sig}}}, E^{\text{den}, \rho_{\text{sig}}}, F^{\rho_{\text{sig}}}$ by their $\mathcal{T}$ -decorated versions.

Corollary 26 (Tree-reduced stored-sign collision-free actual-input sampled Gram). *Assume the hypotheses of the stored-sign collision-free actual-input corollary, and assume the tree-depth roundoff guards above. Let*

$$\hat{G}_{\text{CF}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = \text{fl}\left((\hat{B}^{\rho_{\text{sig}}, \mathcal{T}})^T \hat{B}^{\rho_{\text{sig}}, \mathcal{T}}\right)$$

be the matrix produced by rounded sign storage, tree-reduced rounded sparse CountSketch apply to the actual input A , the concrete denominator $\hat{d}_\star$, rounded sampled-row divisions, and rounded length- s Gram dot products. Then

$$\mathbb{P}_{h, \omega, \sigma} \left[-\varepsilon A^T A - T_{\text{CF}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} I_n \preceq \hat{G}_{\text{CF}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} - A^T A \preceq \varepsilon A^T A + T_{\text{CF}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} I_n \right] \geq 1 - \delta.$$

Exact objects are $U, C, A = UC$, the hash/sign/row-sampling laws, the tree shapes, $Y = S_{h, \omega} A$, and $A^T A$. Computed non-probability objects are the stored signs, tree-reduced sparse products and accumulations, $\hat{d}_\star$, row divisions, and Gram dot products. No extra probability loss is introduced by the tree choice.

For fixed realized $h, \omega, \sigma, A, \mathcal{T}$, fixed $d > 0$, and nonzero leading deterministic coefficient,

$$\begin{aligned}T_{\text{CF}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} &= \Theta\left(\gamma_s \left\| |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}|^T |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}| \right\|_F + \left\| (E^{\text{den}, \rho_{\text{sig}}, \mathcal{T}})^T |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}| + |\tilde{B}^{\rho_{\text{sig}}, \mathcal{T}}|^T E^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} \right\|_F \right. \\ &\quad \left. + \left\| (F^{\rho_{\text{sig}}, \mathcal{T}})^T |\tilde{B}^{\rho_{\text{sig}}, \mathcal{T}}| + |B|^T F^{\rho_{\text{sig}}, \mathcal{T}} \right\|_F \right),\end{aligned}$$

with

$$E_{tj}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} = \Theta\left(u|\hat{B}_{tj}^{\rho_{\text{sig}}, \mathcal{T}}| + \frac{|\hat{Y}_{\sigma_{tj}}^{\rho_{\text{sig}}, \mathcal{T}}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u}\right)$$

and

$$F_{tj}^{\rho_{\text{sig}}, \mathcal{T}} = \Theta\left(\frac{u + \gamma_{d_{\sigma_t}^T(h)}}{\sqrt{s/r}} \sum_{k: h(k) = \sigma_t} |A_{kj}|\right).$$

If the bucket trees are balanced, then $\gamma_{d_i^T(h)} = \Theta(u \log(b_i(h) + 1))$ in the small-roundoff fixed-realization regime, so the sampled sparse-apply term is

$$F_{tj}^{\rho_{\text{sig}}, \mathcal{T}} = \Theta \left(\frac{u(1 + \log(b_{\sigma_t}(h) + 1))}{\sqrt{s/r}} \sum_{k:h(k)=\sigma_t} |A_{kj}| \right).$$

The probability loss remains

$$\Theta \left(\frac{m^2}{r} + \Phi(\theta, r, \varepsilon, s, q) \right),$$

and the deterministic radius vanishes as

$$u, \gamma_s, \gamma_{d_i^T(h)}, \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0$$

for fixed realized data.

Product-Grid Stored CountSketch Signs

If the realized CountSketch signs are stored before the sparse apply, the probability law is still exact, but the stored sign table is a computed non-probability object. For a storage mode

$$\rho \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

define

$$\widehat{\omega}_k^\rho \in \{\text{fl}(\omega_k \cdot 1), \text{fl}(\omega_k + 0), \text{fl}(\omega_k - 0)\}.$$

Since $|\omega_k| = 1$, each of these concrete storage modes satisfies

$$|\widehat{\omega}_k^\rho - \omega_k| \leq u.$$

The computed sparse apply now multiplies A_{kj} by $\widehat{\omega}_k^\rho$, not by the exact sign. Its irreducible entrywise radius is

$$E_{ij}^{\text{CS}, \rho}(h, \omega; A) := (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{k:h(k)=i} |\widehat{\omega}_k^\rho| |A_{kj}| + u \sum_{k:h(k)=i} |A_{kj}|.$$

The first term is the sparse product and bucket-accumulation error using the stored signs; the second term is the exact deterministic storage deviation back to the ideal $S_{h, \omega} A$. The resulting deterministic Gram radius is obtained from the previous display by replacing E^{CS} with $E^{\text{CS}, \rho}$:

$$\begin{aligned} T_{\text{CSGram}, \rho}^{\text{fp}}(h, \omega; A) := & \left\| \left(\gamma_r \sum_{i=1}^r (|Y_{ij}| + E_{ij}^{\text{CS}, \rho}) (|Y_{ik}| + E_{ik}^{\text{CS}, \rho}) \right)_{j,k=1}^n \right\|_F^n \\ & + \left\| \left(\sum_{i=1}^r [E_{ij}^{\text{CS}, \rho} |Y_{ik}| + |Y_{ij}| E_{ik}^{\text{CS}, \rho} + E_{ij}^{\text{CS}, \rho} E_{ik}^{\text{CS}, \rho}] \right)_{j,k=1}^n \right\|_F^n. \end{aligned}$$

For each fixed realized h, ω, A ,

$$\|\widehat{G}^\rho - Y^T Y\|_F \leq T_{\text{CSGram}, \rho}^{\text{fp}}(h, \omega; A).$$

Moreover

$$T_{\text{CSGram},\rho}^{\text{fp}} = \Theta(\gamma_r \| |Y|^T |Y| \|_F + \|(E^{\text{CS},\rho})^T |Y| + |Y|^T E^{\text{CS},\rho}\|_F),$$

with

$$E_{ij}^{\text{CS},\rho} = \Theta\left((u + \gamma_{b_i(h)}) \sum_{k:h(k)=i} |A_{kj}|\right)$$

for the three displayed storage modes. Thus the stored-sign radius is non-vacuous: it tends to zero as $u, \gamma_r, \gamma_{b_i} \rightarrow 0$ for fixed realized data and fixed A .

Corollary 27 (Product-grid CountSketch sparse-Gram theorem with stored signs). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix handed to Algorithm 3, and let the exact CountSketch law draw h and ω . Let $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$ select one of the three sign-storage routines above. Let $\mathcal{I}$ be a finite exact index set, let $g : \mathcal{I} \rightarrow \mathbb{R}$ be an exact one-dimensional grid covering $[-1, 1]$ with radius δ_{grid} , and assume*

$$\delta_{\text{grid}} \geq 0, \quad \sqrt{n} \delta_{\text{grid}} \leq \rho, \quad \eta > 0, \quad L > 0.$$

For every $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, define

$$z_a(j) = g(a(j)), \quad j = 1, \dots, n.$$

Set

$$\tau_{\text{pg},\rho_{\text{sig}}}^{\text{fp}}(h, \omega; A) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram},\rho_{\text{sig}}}^{\text{fp}}(h, \omega; A).$$

If the floating-point model has valid γ_m and γ_r , then

$$\begin{aligned} \mathbb{P}_{h,\omega} \left[-\tau_{\text{pg},\rho_{\text{sig}}}^{\text{fp}}(h, \omega; A) I_n \preceq \widehat{G}^{\rho_{\text{sig}}}(h, \omega; A) - A^T A \preceq \tau_{\text{pg},\rho_{\text{sig}}}^{\text{fp}}(h, \omega; A) I_n \right] \\ \geq 1 - \left(\sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \frac{2 \|Az_a\|_2^4}{r\eta^2} + \frac{2 \|A\|_F^4}{rL^2} \right). \end{aligned}$$

Equivalently, if the displayed product-grid loss is at most δ , the stored-sign computed sparse-Gram Loewner event has probability at least $1 - \delta$.

The product-grid loss is unchanged by sign storage because it belongs to the exact hash/sign probability analysis. The computed non-probability quantities are the stored sign table $\widehat{\omega}^{\rho_{\text{sig}}}$, the sparse signed products, bucket accumulations, and rounded Gram dot products; their floating-point contribution is exactly the deterministic radius $T_{\text{CSGram},\rho_{\text{sig}}}^{\text{fp}}$. For fixed exact A , fixed grid, fixed η, L , and nonzero numerator, the probability-loss order is

$$\Theta\left(\frac{\sum_a \|Az_a\|_2^4 / \eta^2 + \|A\|_F^4 / L^2}{r}\right),$$

while the additional floating-point radius has the Θ -order displayed above and tends to zero with $u, \gamma_r, \gamma_{b_i}$.

Stored CountSketch Signs with Permuted Bucket Layouts

The preceding stored-sign theorem used the canonical exact enumeration of each bucket. A sparse implementation may instead visit the selected rows in a different fixed memory order. For each realized exact hash h , let

$$B_i(h) = \{k : h(k) = i\}, \quad b_i(h) = |B_i(h)|,$$

and let

$$k_{i,1}, \dots, k_{i,b_i(h)}$$

be any exact ordering of the same bucket. This ordering is a discrete index operation, not a floating-point real computation. With stored signs $\hat{\omega}^\rho$, define the permuted sparse-apply radius

$$\begin{aligned} E_{ij}^{\text{CS},\rho,\pi}(h, \omega; A) &:= (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{t=1}^{b_i(h)} |\hat{\omega}_{k_{i,t}}^{\rho_{\text{sig}}}||A_{k_{i,t},j}| \\ &\quad + u \sum_{t=1}^{b_i(h)} |A_{k_{i,t},j}|. \end{aligned}$$

The first line charges the rounded signed products and the rounded left-to-right accumulation in the chosen order. The second line charges the storage error of the realized signs. The exact ideal reference remains

$$Y_{ij} = \sum_{k:h(k)=i} \omega_k A_{kj}.$$

The corresponding Gram radius is

$$\begin{aligned} T_{\text{CSGram},\rho,\pi}^{\text{fp}}(h, \omega; A) &:= \left\| \left(\gamma_r \sum_{i=1}^r (|Y_{ij}| + E_{ij}^{\text{CS},\rho,\pi})(|Y_{ik}| + E_{ik}^{\text{CS},\rho,\pi}) \right)_{j,k=1}^n \right\|_F^n \\ &\quad + \left\| \left(\sum_{i=1}^r [E_{ij}^{\text{CS},\rho,\pi}|Y_{ik}| + |Y_{ij}|E_{ik}^{\text{CS},\rho,\pi} + E_{ij}^{\text{CS},\rho,\pi}E_{ik}^{\text{CS},\rho,\pi}] \right)_{j,k=1}^n \right\|_F^n. \end{aligned}$$

Corollary 28 (Product-grid CountSketch sparse-Gram theorem with stored signs and permuted buckets). *Keep the product-grid hypotheses of the preceding stored-sign corollary, and fix any exact bucket ordering π for every realized hash h . Let*

$$\hat{G}^{\rho_{\text{sig}},\pi}(h, \omega; A)$$

be the computed Gram formed by storing the signs with mode ρ_{sig} , visiting each bucket in the chosen order, accumulating the sparse products in that order, and then forming rounded length- r Gram dot products. Set

$$\tau_{\text{pg},\rho_{\text{sig}},\pi}^{\text{fp}}(h, \omega; A) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram},\rho_{\text{sig}},\pi}^{\text{fp}}(h, \omega; A).$$

If the floating-point model has valid γ_m and γ_r , then

$$\begin{aligned} \mathbb{P}_{h,\omega} \left[-\tau_{\text{pg},\rho_{\text{sig}},\pi}^{\text{fp}}(h, \omega; A) I_n \preceq \hat{G}^{\rho_{\text{sig}},\pi}(h, \omega; A) - A^T A \preceq \tau_{\text{pg},\rho_{\text{sig}},\pi}^{\text{fp}}(h, \omega; A) I_n \right] \\ \geq 1 - \left(\sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \frac{2\|Az_a\|_2^4}{r\eta^2} + \frac{2\|A\|_F^4}{rL^2} \right). \end{aligned}$$

Equivalently, the same event has probability at least $1 - \delta$ whenever the displayed product-grid loss is at most δ .

The exact objects are A , h , ω , the bucket order π , the grid vectors z_a , and the reference $A^T A$. The computed non-probability objects are the stored sign table, the rounded signed products, the

rounded bucket accumulations in the chosen order, and the rounded Gram dot products. For fixed realized h, ω, π, A ,

$$E_{ij}^{\text{CS}, \rho, \pi} = \Theta \left((u + \gamma_{b_i(h)}) \sum_{t=1}^{b_i(h)} |A_{k_{i,t}j}| \right)$$

for the three stored-sign modes, and

$$T_{\text{CSGram}, \rho, \pi}^{\text{fp}} = \Theta(\gamma_r \| |Y|^T |Y| \|_F + \| (E^{\text{CS}, \rho, \pi})^T |Y| + |Y|^T E^{\text{CS}, \rho, \pi} \|_F)$$

when the displayed leading term is nonzero. The roundoff radius tends to zero as $u, \gamma_r, \gamma_{b_i} \rightarrow 0$ for fixed realized data and fixed A .

Stored CountSketch Signs with Tree-Reduced Bucket Accumulation

The preceding stored-sign and permuted-bucket statements use a left-to-right fold within each CountSketch bucket. A parallel sparse implementation may instead reduce the bucket entries by a binary summation tree. For every realized exact hash h , define

$$B_i(h) = \{k : h(k) = i\}, \quad b_i(h) = |B_i(h)|,$$

and fix an exact enumeration

$$k_{i,1}, \dots, k_{i,b_i(h)}$$

of $B_i(h)$. For each bucket i , let

$$\mathcal{T}_i(h)$$

be an exact binary summation tree with $b_i(h) + 1$ leaves. The first $b_i(h)$ leaves hold the rounded signed products, and the last leaf is the exact zero padding value. The zero padding is not a floating-point operation; it is only a modeling convention that lets empty buckets use the same tree statement. Let

$$d_i(h) = \text{depth}(\mathcal{T}_i(h)).$$

The floating-point assumptions for this route are valid $\gamma_{d_i(h)}$ for every realized bucket tree and valid γ_r for each length- r Gram dot product.

As before, the exact Rademacher signs are $\omega_k \in \{-1, 1\}$. The implemented sign table is

$$\hat{\omega}^{\rho_{\text{sig}}}, \quad \rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

where the three modes store signs by $\text{fl}(\omega_k \cdot 1)$, $\text{fl}(\omega_k + 0)$, or $\text{fl}(\omega_k - 0)$. Let $e_k^{\rho_{\text{sig}}}$ be the certified storage radius satisfying

$$|\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| \leq e_k^{\rho_{\text{sig}}}.$$

For the three concrete modes above, $e_k^{\rho_{\text{sig}}} \leq u$ because $|\omega_k| = 1$. Define the tree-reduced stored-sign sparse-apply entry radius

$$\begin{aligned} E_{ij}^{\text{CS}, \rho, \mathcal{T}}(h, \omega; A) &:= (u + \gamma_{d_i(h)} + u\gamma_{d_i(h)}) \sum_{t=1}^{b_i(h)} |\hat{\omega}_{k_{i,t}}^{\rho_{\text{sig}}}||A_{k_{i,t}j}| \\ &\quad + \sum_{t=1}^{b_i(h)} e_{k_{i,t}}^{\rho_{\text{sig}}} |A_{k_{i,t}j}|. \end{aligned}$$

The first line charges the rounded signed products and the rounded tree reduction. The second line charges the stored sign table. The exact ideal reference is still

$$Y_{ij} = \sum_{k:h(k)=i} \omega_k A_{kj}.$$

Let $\hat{Y}^{\rho_{\text{sig}}, \mathcal{T}}$ be the computed sparse sketch formed from the stored signs, rounded signed products, and the bucket trees. The corresponding fully expanded deterministic Gram radius is

$$\begin{aligned} T_{\text{CSGram}, \rho, \mathcal{T}}^{\text{fp}}(h, \omega; A) := & \left\| \left(\gamma_r \sum_{i=1}^r (|Y_{ij}| + E_{ij}^{\text{CS}, \rho, \mathcal{T}}) (|Y_{ik}| + E_{ik}^{\text{CS}, \rho, \mathcal{T}}) \right)_{j,k=1}^n \right\|_F^n \\ & + \left\| \left(\sum_{i=1}^r [E_{ij}^{\text{CS}, \rho, \mathcal{T}} |Y_{ik}| + |Y_{ij}| E_{ik}^{\text{CS}, \rho, \mathcal{T}} + E_{ij}^{\text{CS}, \rho, \mathcal{T}} E_{ik}^{\text{CS}, \rho, \mathcal{T}}] \right)_{j,k=1}^n \right\|_F^n. \end{aligned}$$

Corollary 29 (Product-grid CountSketch sparse-Gram theorem with stored signs and tree reductions). *Keep the product-grid hypotheses of the stored-sign corollary, and fix exact bucket trees $\mathcal{T}_i(h)$ for every realized hash h . Let*

$$\hat{G}^{\rho_{\text{sig}}, \mathcal{T}}(h, \omega; A) = \text{fl} \left((\hat{Y}^{\rho_{\text{sig}}, \mathcal{T}})^T \hat{Y}^{\rho_{\text{sig}}, \mathcal{T}} \right)$$

be the computed Gram whose length- r dot products are rounded. Set

$$\tau_{\text{pg}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}(h, \omega; A) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}(h, \omega; A).$$

If the floating-point model has valid $\gamma_{d_i(h)}$ for every bucket tree and valid γ_r , then

$$\begin{aligned} \mathbb{P}_{h, \omega} \left[-\tau_{\text{pg}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}(h, \omega; A) I_n \preceq \hat{G}^{\rho_{\text{sig}}, \mathcal{T}}(h, \omega; A) - A^T A \preceq \tau_{\text{pg}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}(h, \omega; A) I_n \right] \\ \geq 1 - \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2 \|Az_a\|_2^4}{r\eta^2} + \frac{2 \|A\|_F^4}{rL^2} \right). \end{aligned}$$

Equivalently, the same event has probability at least $1 - \delta$ whenever the displayed product-grid loss is at most δ .

The exact objects are A, h, ω , the product-grid vectors z_a , the reference $A^T A$, and the tree shapes $\mathcal{T}_i(h)$. These tree shapes are exact discrete algorithm choices, not floating-point real computations. The computed non-probability objects are the stored sign table, the rounded signed products, the rounded tree-reduced bucket accumulations, and the rounded Gram dot products.

For fixed realized $h, \omega, \mathcal{T}, A$, and for the three concrete stored-sign modes, the entry budget has small-roundoff order

$$E_{ij}^{\text{CS}, \rho, \mathcal{T}} = \Theta \left((u + \gamma_{d_i(h)}) \sum_{t=1}^{b_i(h)} |A_{k_{i,t}j}| \right)$$

whenever the displayed bucket-column absolute sum is nonzero. If balanced trees are selected, so that

$$d_i(h) = \Theta(\log(b_i(h) + 1)),$$

then $\gamma_{d_i(h)} = \Theta(u \log(b_i(h) + 1))$ in the usual small- u regime, and therefore

$$E_{ij}^{\text{CS}, \rho, \mathcal{T}} = \Theta \left(u(1 + \log(b_i(h) + 1)) \sum_{t=1}^{b_i(h)} |A_{k_{i,t}j}| \right).$$

The deterministic Gram radius satisfies

$$T_{\text{CSGram}, \rho, \mathcal{T}}^{\text{fp}} = \Theta(\gamma_r \| |Y|^T |Y| \|_F + \|(E^{\text{CS}, \rho, \mathcal{T}})^T |Y| + |Y|^T E^{\text{CS}, \rho, \mathcal{T}}\|_F)$$

when the displayed leading term is nonzero; the exact expanded formula above retains the higher-order $E^T E$ and $\gamma_r E$ -dependent terms. The probability-loss order is unchanged,

$$\Theta \left(\frac{\sum_a \|Az_a\|_2^4 / \eta^2 + \|A\|_F^4 / L^2}{r} \right)$$

for fixed exact input/grid/threshold data and nonzero numerator. The result is non-vacuous whenever the displayed product-grid loss is below one, and the deterministic roundoff radius tends to zero as

$$u, \gamma_r, \gamma_{d_i(h)}, e_k^{\rho_{\text{sig}}} \longrightarrow 0$$

for fixed realized data and fixed exact A .

Corollary 30 (Product-grid CountSketch sparse-Gram theorem). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix handed to Algorithm 3. Let $\mathcal{I}$ be a finite exact index set, and let*

$$g : \mathcal{I} \rightarrow \mathbb{R}$$

be an exact one-dimensional grid that covers $[-1, 1]$ with radius δ_{grid} . Assume

$$\delta_{\text{grid}} \geq 0, \quad \sqrt{n} \delta_{\text{grid}} \leq \rho, \quad \eta > 0, \quad L > 0.$$

For each product-grid index $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, define

$$z_a(j) = g(a(j)), \quad j = 1, \dots, n.$$

Then $\{z_a\}$ is an exact ρ -cover of the Euclidean unit ball in $\mathbb{R}^n$, and it has cardinality $|\mathcal{I}|^n$.

Set

$$\tau_{\text{pg}}^{\text{fp}}(h, \omega; A) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram}}^{\text{fp}}(h, \omega; A).$$

If the floating-point model has valid γ_m and γ_r , then

$$\begin{aligned} \mathbb{P}_{h, \omega} \left[-\tau_{\text{pg}}^{\text{fp}}(h, \omega; A) I_n \preceq \widehat{G}(h, \omega; A) - A^T A \preceq \tau_{\text{pg}}^{\text{fp}}(h, \omega; A) I_n \right] \\ \geq 1 - \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2 \|Az_a\|_2^4}{r \eta^2} + \frac{2 \|A\|_F^4}{r L^2} \right). \end{aligned}$$

Equivalently, if the displayed product-grid loss is at most δ , the same computed sparse-Gram Loewner event has probability at least $1 - \delta$.

All quantities in the product-grid loss are exact arithmetic analysis quantities: A , the grid g , the product vectors z_a , and the exact hash/sign law. The computed non-probability quantities are precisely $\widehat{Y}$ and $\widehat{G}$, and their floating-point errors are charged by the expanded $T_{\text{CSGram}}^{\text{fp}}$ above.

For fixed exact A , fixed grid g , and fixed positive η, L , the probability loss has order

$$\Theta\left(\frac{\sum_{a:\{1,\dots,n\}\rightarrow\mathcal{I}} \|Az_a\|_2^4/\eta^2 + \|A\|_F^4/L^2}{r}\right)$$

when the numerator is nonzero, and it is exactly zero when the numerator is zero. The probability bound is non-vacuous whenever the displayed loss is less than one. The floating-point radius is non-vacuous because

$$T_{\text{CSGram}}^{\text{fp}}(h, \omega; A) \rightarrow 0 \quad \text{as} \quad u, \gamma_r, \gamma_{b_i(h)} \rightarrow 0$$

for fixed realized h, ω and fixed exact input A .

Corollary 31 (Orthonormal product-grid CountSketch sparse-Gram theorem). *Keep the exact CountSketch product-grid notation above. Let $U \in \mathbb{R}^{m \times n}$ be an exact orthonormal-column analysis matrix,*

$$U^T U = I_n.$$

For each product-grid index $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, define the exact grid vector

$$z_a(j) = g(a(j)).$$

Because U has orthonormal columns,

$$\|U z_a\|_2^2 = \|z_a\|_2^2, \quad \|U\|_F^2 = n.$$

Thus the exact product-grid probability loss reduces to the irreducible orthonormal form

$$\mathcal{L}_{\text{pg}}^{\text{orth}} = \sum_{a:\{1,\dots,n\}\rightarrow\mathcal{I}} \frac{2\|z_a\|_2^4}{r\eta^2} + \frac{2n^2}{rL^2}.$$

Let

$$\tau_{\text{pg}}^{\text{fp}}(h, \omega; U) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram}}^{\text{fp}}(h, \omega; U),$$

where $T_{\text{CSGram}}^{\text{fp}}$ is the sparse-apply and rounded-Gram radius defined above. If the floating-point model has valid γ_m and γ_r , then

$$\begin{aligned} \mathbb{P}_{h,\omega} \left[-\tau_{\text{pg}}^{\text{fp}}(h, \omega; U) I_n \preceq \widehat{G}(h, \omega; U) - I_n \preceq \tau_{\text{pg}}^{\text{fp}}(h, \omega; U) I_n \right] \\ \geq 1 - \mathcal{L}_{\text{pg}}^{\text{orth}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg}}^{\text{orth}} \leq \delta$, the same computed sparse-Gram Loewner event has probability at least $1 - \delta$.

For the stored-sign implementation path, let $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$, store the realized signs as $\widehat{\omega}^{\rho_{\text{sig}}}$, and replace $T_{\text{CSGram}}^{\text{fp}}$ by $T_{\text{CSGram}, \rho_{\text{sig}}}^{\text{fp}}$. Then the same probability loss $\mathcal{L}_{\text{pg}}^{\text{orth}}$ holds for the stored-sign computed event

$$-\tau_{\text{pg}, \rho_{\text{sig}}}^{\text{fp}}(h, \omega; U) I_n \preceq \widehat{G}^{\rho_{\text{sig}}}(h, \omega; U) - I_n \preceq \tau_{\text{pg}, \rho_{\text{sig}}}^{\text{fp}}(h, \omega; U) I_n,$$

where

$$\tau_{\text{pg}, \rho_{\text{sig}}}^{\text{fp}}(h, \omega; U) = \eta + L(2\rho + \rho^2) + T_{\text{CSGram}, \rho_{\text{sig}}}^{\text{fp}}(h, \omega; U).$$

The exact objects are U , the grid g , the product-grid vectors z_a , and the hash/sign probability law. In the stored-sign path, the computed non-probability objects are the stored sign table, sparse signed

products, bucket accumulations, and rounded Gram dot products. The statement does not charge probability construction, and it does not rely on an auxiliary perturbation event or supplied certificate.

For fixed exact grid and fixed positive η, L , the irreducible probability-loss order is

$$\mathcal{L}_{\text{pg}}^{\text{orth}} = \Theta \left(\frac{\sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \|z_a\|_2^4 / \eta^2 + n^2 / L^2}{r} \right)$$

when the numerator is nonzero, and the loss is exactly zero when the numerator is zero. The result is non-vacuous whenever $\mathcal{L}_{\text{pg}}^{\text{orth}} < 1$. The deterministic floating-point radius tends to zero as the relevant $u, \gamma_r, \gamma_{b_i(h)}$, and stored-sign copy radii tend to zero for fixed realized data.

Downstream Uniform Rows After CountSketch

After the CountSketch preprocessing, Algorithm 3 may sample s rows from the r -row sketched matrix. Let

$$\sigma = (\sigma_1, \dots, \sigma_s), \quad \sigma_q \in \{1, \dots, r\},$$

be exact iid uniform row indices. Define the exact denominator

$$d = \sqrt{s/r}.$$

The concrete denominator routine used here computes

$$\hat{d} = \text{fl} \left(\sqrt{s} \text{fl} \left(\frac{1}{\sqrt{r}} \right) \right),$$

implemented as

$$\text{fl}_{\text{mul}}(\text{fl}_{\text{sqr}}(s), \text{fl}_{\text{div}}(1, \text{fl}_{\text{sqr}}(r))),$$

and the proved denominator error radius is

$$E_d = d \frac{4u + 3u^2 + u^3}{1 - u}, \quad |\hat{d} - d| \leq E_d.$$

Assume $\hat{d} \neq 0$, which is part of the concrete denominator certificate.

For a realized h, ω, σ , define the exact scaled sampled rows

$$B_{qj} = \frac{Y_{\sigma_{qj}}}{d}, \quad q = 1, \dots, s, \quad j = 1, \dots, n.$$

The implementation uses the already computed sparse sketch $\hat{Y}$ and forms

$$\hat{B}_{qj} = \text{fl} \left(\frac{\hat{Y}_{\sigma_{qj}}}{\hat{d}} \right).$$

An explicit entrywise radius for this sampled-row scaling is

$$E_{qj}^B = \frac{u|\hat{Y}_{\sigma_{qj}}|}{|\hat{d}|} + \frac{E_{\sigma_{qj}}^{\text{CS}}}{|\hat{d}|} + \frac{|Y_{\sigma_{qj}}|E_d}{|\hat{d}||d|}.$$

The first term charges the rounded division by $\hat{d}$, the second term propagates the computed sparse-apply error, and the third term charges the computed denominator.

The final sampled Gram is

$$\widehat{G}_{\text{row}}(h, \omega, \sigma; A) = \text{fl}(\widehat{B}^T \widehat{B}),$$

with rounded length- s dot products. Its fully expanded deterministic floating-point radius is

$$\begin{aligned} T_{\text{CS}, \text{row}}^{\text{fp}} := & \left\| \left(\gamma_s \sum_{q=1}^s (|B_{qj}| + E_{qj}^B)(|B_{qk}| + E_{qk}^B) \right)_{j,k=1}^n \right\|_F^n \\ & + \left\| \left(\sum_{q=1}^s [E_{qj}^B |B_{qk}| + |B_{qj}| E_{qk}^B + E_{qj}^B E_{qk}^B] \right)_{j,k=1}^n \right\|_F^n. \end{aligned}$$

For fixed realized h, ω, σ, A , this radius tends to zero as

$$u, \gamma_s, \gamma_{b_i(h)}, E_d \longrightarrow 0.$$

Corollary 32 (Product-grid CountSketch plus downstream uniform rows). *Keep the hypotheses and product-grid notation of the preceding CountSketch corollary. Let $\eta_{\text{row}} > 0$, and set*

$$\tau_{\text{CS}} = \eta + L(2\rho + \rho^2).$$

Then the fully computed downstream sampled Gram satisfies

$$\begin{aligned} \mathbb{P}_{h, \omega, \sigma} \left[-(\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}}(h, \omega, \sigma; A) - A^T A \right. \\ \left. \preceq (\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}}^{\text{fp}}) I_n \right] \\ \geq 1 - \mathcal{L}_{\text{pg}, \text{row}}, \end{aligned}$$

where the exact product-grid loss is

$$\begin{aligned} \mathcal{L}_{\text{pg}, \text{row}} := & \sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{2}{r\eta^2} \sum_{p \neq q} ((Az_a)_p (Az_a)_q)^2 \\ & + \frac{2}{rL^2} \sum_{j,k=1}^n \sum_{p \neq q} (A_{pj} A_{qk})^2 + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg}, \text{row}} \leq \delta$, then the displayed fully computed event has probability at least $1 - \delta$.

The exact objects are A , the product grid z_a , the exact hash/sign law, and the exact uniform row-sampling law. The computed non-probability objects are $\widehat{Y}$, $\widehat{d}$, $\widehat{B}$, and $\widehat{G}_{\text{row}}$. Their errors are charged by E^{CS} , E_d , E^B , and the expanded $T_{\text{CS}, \text{row}}^{\text{fp}}$.

In the fixed exact-input/grid/threshold regime, the probability loss has order

$$\Theta \left(\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \frac{\sum_{p \neq q} ((Az_a)_p (Az_a)_q)^2}{r\eta^2} + \frac{\sum_{j,k=1}^n \sum_{p \neq q} (A_{pj} A_{qk})^2}{rL^2} + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2} \right)$$

when the displayed numerator is nonzero. The result is non-vacuous whenever $\mathcal{L}_{\text{pg}, \text{row}} < 1$, and its deterministic floating-point radius vanishes when the charged roundoff and denominator errors vanish.

Corollary 33 (Product-grid CountSketch plus downstream uniform rows with stored signs). *Keep the exact product-grid notation above, and let the CountSketch signs be stored by one of the concrete routines*

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}.$$

Thus $\widehat{\omega}^{\rho_{\text{sig}}}$ is respectively formed by $\text{fl}(\omega_k \cdot 1)$, $\text{fl}(\omega_k + 0)$, or $\text{fl}(\omega_k - 0)$, and

$$E_{ij}^{\text{CS}, \rho_{\text{sig}}} = (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{k:h(k)=i} |\widehat{\omega}_k^{\rho_{\text{sig}}}| |A_{kj}| + u \sum_{k:h(k)=i} |A_{kj}|$$

is the stored-sign sparse-apply entry radius defined before the sparse-Gram corollary.

For a realized row trace $\sigma = (\sigma_1, \dots, \sigma_s)$, define

$$\widehat{Y}^{\rho_{\text{sig}}} = \text{fl}(S_{h, \widehat{\omega}^{\rho_{\text{sig}}}} A), \quad \widetilde{B}_{qj}^{\rho_{\text{sig}}} = \frac{\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}}}{d}, \quad \widehat{B}_{qj}^{\rho_{\text{sig}}} = \text{fl}\left(\frac{\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}}}{\widehat{d}}\right).$$

The exact reference $B_{qj} = Y_{\sigma_q j}/d$ uses the exact CountSketch signs and the exact denominator. The computed-denominator row entry radius for the stored-sign computed sketch is

$$E_{qj}^{\text{den}, \rho_{\text{sig}}} = |\widehat{B}_{qj}^{\rho_{\text{sig}}}| u + \frac{|\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}}| E_d}{|\widehat{d}| |d|}.$$

The stored-sign basis perturbation sampled at row q is

$$F_{qj}^{\rho_{\text{sig}}} = \frac{E_{\sigma_q j}^{\text{CS}, \rho_{\text{sig}}}}{d}.$$

Now define the irreducible deterministic downstream radius

$$T_{\text{CS}, \text{row}, \rho_{\text{sig}}}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}}^{\text{fp}},$$

where

$$T_{\text{den}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\sum_{q=1}^s E_{qj}^{\text{den}, \rho_{\text{sig}}} |\widehat{B}_{qk}^{\rho_{\text{sig}}}| + |\widetilde{B}_{qj}^{\rho_{\text{sig}}}| E_{qk}^{\text{den}, \rho_{\text{sig}}} \right)_{j,k=1}^n \right\|_F,$$

$$T_{\text{dot}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\gamma_s \sum_{q=1}^s |\widehat{B}_{qj}^{\rho_{\text{sig}}}| |\widehat{B}_{qk}^{\rho_{\text{sig}}}| \right)_{j,k=1}^n \right\|_F,$$

and

$$T_{\text{basis}, \rho_{\text{sig}}}^{\text{fp}} = \left\| \left(\sum_{q=1}^s F_{qj}^{\rho_{\text{sig}}} |\widetilde{B}_{qk}^{\rho_{\text{sig}}}| + |B_{qj}| F_{qk}^{\rho_{\text{sig}}} \right)_{j,k=1}^n \right\|_F.$$

With $\eta_{\text{row}} > 0$ and

$$\tau_{\text{CS}} = \eta + L(2\rho + \rho^2),$$

the fully computed stored-sign downstream sampled Gram

$$\widehat{G}_{\text{row}}^{\rho_{\text{sig}}}(h, \omega, \sigma; A) = \text{fl}\left((\widehat{B}^{\rho_{\text{sig}}})^T \widehat{B}^{\rho_{\text{sig}}}\right)$$

satisfies

$$\begin{aligned} \mathbb{P}_{h,\omega,\sigma} \left[-(\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS,row},\rho_{\text{sig}}}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}}^{\rho_{\text{sig}}}(h, \omega, \sigma; A) - A^T A \right. \\ \left. \preceq (\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS,row},\rho_{\text{sig}}}^{\text{fp}}) I_n \right] \\ \geq 1 - \mathcal{L}_{\text{pg,row}}. \end{aligned}$$

Equivalently, if the exact loss $\mathcal{L}_{\text{pg,row}}$ displayed in the preceding corollary is at most δ , then this stored-sign fully computed event has probability at least $1 - \delta$.

The exact objects are A , the product grid z_a , the exact CountSketch hash/sign law, and the exact uniform row-sampling law. The computed non-probability objects are $\widehat{\omega}^{\rho_{\text{sig}}}$, $\widehat{Y}^{\rho_{\text{sig}}}$, $\widehat{d}$, $\widehat{B}^{\rho_{\text{sig}}}$, and $\widehat{G}_{\text{row}}^{\rho_{\text{sig}}}$. No perturbation event or extra existence assumption is supplied.

For fixed realized h, ω, σ, A , fixed exact denominator $d > 0$, and nonzero leading deterministic coefficient,

$$\begin{aligned} T_{\text{CS,row},\rho_{\text{sig}}}^{\text{fp}} = \Theta \Big(\gamma_s \left\| |\widehat{B}^{\rho_{\text{sig}}}|^T |\widehat{B}^{\rho_{\text{sig}}}| \right\|_F \\ + \left\| (E^{\text{den},\rho_{\text{sig}}})^T |\widehat{B}^{\rho_{\text{sig}}}| + |\widetilde{B}^{\rho_{\text{sig}}}|^T E^{\text{den},\rho_{\text{sig}}} \right\|_F \\ + \left\| (F^{\rho_{\text{sig}}})^T |\widetilde{B}^{\rho_{\text{sig}}}| + |B|^T F^{\rho_{\text{sig}}} \right\|_F \Big). \end{aligned}$$

Substituting the irreducible definitions,

$$E_{qj}^{\text{den},\rho_{\text{sig}}} = \Theta \left(u |\widehat{B}_{qj}^{\rho_{\text{sig}}}| + \frac{|\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}}|}{|\widehat{d}| |d|} d \frac{4u + 3u^2 + u^3}{1 - u} \right),$$

and

$$F_{qj}^{\rho_{\text{sig}}} = \Theta \left(\frac{u + \gamma_{b_{\sigma_q}(h)}}{d} \sum_{k:h(k)=\sigma_q} |A_{kj}| \right).$$

Thus, in the small-roundoff regime with fixed realized data,

$$T_{\text{CS,row},\rho_{\text{sig}}}^{\text{fp}} \rightarrow 0 \quad \text{as} \quad u, \gamma_s, \gamma_{b_i}(h), \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0.$$

The probability-loss order remains the same Θ -order as in the preceding downstream corollary, because sign storage is a deterministic floating-point radius and not a probability-law perturbation.

Corollary 34 (Orthonormal product-grid CountSketch plus downstream uniform rows with stored signs). *Keep the product-grid, stored-sign, and downstream row notation introduced above. Now specialize the exact input to an orthonormal-column analysis matrix $U \in \mathbb{R}^{m \times n}$, so*

$$U^T U = I_n.$$

For each product-grid index $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, keep

$$z_a(j) = g(a(j)).$$

The orthonormal identities

$$\|U z_a\|_2^2 = \|z_a\|_2^2, \quad \|U\|_F^2 = n$$

turn the stored-sign downstream product-grid loss into

$$\mathcal{L}_{\text{pg,row}}^{\text{orth}} := \sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \frac{2 \|z_a\|_2^4}{r \eta^2} + \frac{2n^2}{r L^2} + \frac{r(mn)^2}{s \eta_{\text{row}}^2}.$$

Let $\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\}$ be one of the three stored-sign copy routines defined above, and let

$$\tau_{\text{pg,row},\rho_{\text{sig}}}^{\text{fp}}(h, \omega, \sigma; U) = \eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS,row},\rho_{\text{sig}}}^{\text{fp}}(h, \omega, \sigma; U).$$

Then the fully computed stored-sign downstream sampled Gram satisfies

$$\begin{aligned} \mathbb{P}_{h,\omega,\sigma} \left[-\tau_{\text{pg,row},\rho_{\text{sig}}}^{\text{fp}}(h, \omega, \sigma; U) I_n \preceq \widehat{G}_{\text{row}}^{\rho_{\text{sig}}}(h, \omega, \sigma; U) - I_n \right. \\ \left. \preceq \tau_{\text{pg,row},\rho_{\text{sig}}}^{\text{fp}}(h, \omega, \sigma; U) I_n \right] \\ \geq 1 - \mathcal{L}_{\text{pg,row}}^{\text{orth}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg,row}}^{\text{orth}} \leq \delta$, the same fully computed event has probability at least $1 - \delta$.

The exact objects are U , the product grid g , the grid vectors z_a , the exact CountSketch hash/sign law, and the exact downstream uniform-row law. The computed non-probability objects are the stored sign table, $\widehat{Y}^{\rho_{\text{sig}}}$, the concrete denominator $\widehat{d}$, the rounded sampled rows $\widehat{B}^{\rho_{\text{sig}}}$, and the rounded sampled Gram $\widehat{G}_{\text{row}}^{\rho_{\text{sig}}}$. The displayed radius charges sign storage, sparse signed products, bucket accumulation, denominator formation, row divisions, and sampled-Gram dot products.

For fixed exact grid and fixed positive $\eta, L, \eta_{\text{row}}$, the irreducible probability-loss order is

$$\mathcal{L}_{\text{pg,row}}^{\text{orth}} = \Theta \left(\frac{\sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \|z_a\|_2^4 / \eta^2 + n^2 / L^2}{r} + \frac{r m^2 n^2}{s \eta_{\text{row}}^2} \right)$$

when the displayed numerator is nonzero. The result is non-vacuous whenever $\mathcal{L}_{\text{pg,row}}^{\text{orth}} < 1$. For fixed realized data, its deterministic floating-point radius tends to zero as the unit roundoff, γ_s , the bucket-accumulation γ -factors, and the denominator error tend to zero.

Corollary 35 (Orthonormal product-grid downstream uniform rows with fixed-order or tree-reduced buckets). *Keep the exact orthonormal input U , product-grid vectors z_a , and loss $\mathcal{L}_{\text{pg,row}}^{\text{orth}}$ from the preceding corollary. For every realized exact hash h , consider either of the following exact discrete layout choices.*

First, a fixed-order layout π specifies, for every bucket i , an ordering of the exact bucket set $B_i(h) = \{k : h(k) = i\}$. Second, a tree-reduced layout $\mathcal{T}$ specifies, for every bucket i , an exact binary summation tree with depth $d_i^{\mathcal{T}}(h)$. The choices π and $\mathcal{T}$ are exact discrete inputs, not floating-point computations. Arithmetic performed along the chosen order or tree is floating-point arithmetic and is charged in the deterministic radius.

Let

$$\ell \in \{\pi, \mathcal{T}\}$$

denote one of these two layouts. For a stored-sign routine

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

write

$$\widehat{Y}^{\rho_{\text{sig}},\ell}(h, \omega; U)$$

for the computed sparse CountSketch basis obtained from U by storing the realized signs, forming rounded signed products, and accumulating each bucket according to layout ℓ . The downstream sampled rows use the concrete computed denominator

$$\widehat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{r}))) ,$$

then round the row divisions and the sampled-Gram dot products. Let the final computed sampled Gram be

$$\widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \ell}(h, \omega, \sigma; U).$$

Define

$$\tau_{\text{pg, row}, \rho_{\text{sig}}, \ell}^{\text{fp}}(h, \omega, \sigma; U) = \eta + L(2\rho + \rho^2) + \eta_{\text{row}} + T_{\text{CS, row}, \rho_{\text{sig}}, \ell}^{\text{fp}}(h, \omega, \sigma; U),$$

where $T_{\text{CS, row}, \rho_{\text{sig}}, \pi}^{\text{fp}}$ charges stored signs, rounded sparse products, fixed-order bucket accumulation, denominator formation, sampled-row divisions, and sampled-Gram dot products, while $T_{\text{CS, row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}$ charges the same operations with the bucket-accumulation γ -factor replaced by the tree-depth factors $\gamma_{d_i^{\mathcal{T}}(h)}$. The tree-reduced statement assumes these tree-depth γ -factors are valid.

Then, for either $\ell = \pi$ or $\ell = \mathcal{T}$,

$$\begin{aligned} \mathbb{P}_{h, \omega, \sigma} \left[-\tau_{\text{pg, row}, \rho_{\text{sig}}, \ell}^{\text{fp}}(h, \omega, \sigma; U) I_n \preceq \widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \ell}(h, \omega, \sigma; U) - I_n \right. \\ \left. \preceq \tau_{\text{pg, row}, \rho_{\text{sig}}, \ell}^{\text{fp}}(h, \omega, \sigma; U) I_n \right] \\ \geq 1 - \mathcal{L}_{\text{pg, row}}^{\text{orth}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg, row}}^{\text{orth}} \leq \delta$, the same fully computed layout-specific event has probability at least $1 - \delta$.

The exact objects are U , the grid g , the vectors z_a , the exact hash/sign laws, the exact downstream row law, and the exact layout choice ℓ . The computed non-probability objects are the stored signs, layout-specific sparse products and bucket accumulations, $\widehat{d}_\star$, the rounded sampled rows, and the rounded sampled Gram. No probability construction is charged, and no auxiliary perturbation event is supplied.

The probability-loss order is the same irreducible order as in the preceding corollary,

$$\mathcal{L}_{\text{pg, row}}^{\text{orth}} = \Theta \left(\frac{\sum_{a: \{1, \dots, n\} \rightarrow \mathcal{I}} \|z_a\|_2^4 / \eta^2 + n^2 / L^2}{r} + \frac{r m^2 n^2}{s \eta_{\text{row}}^2} \right),$$

for fixed exact grid and fixed positive $\eta, L, \eta_{\text{row}}$ when the displayed numerator is nonzero. For fixed realized data, the layout-specific deterministic radius tends to zero as the unit roundoff, γ_s , the fixed-order bucket γ -factors, or the tree-depth γ -factors, and the denominator error tend to zero.

Corollary 36 (Product-grid CountSketch plus downstream uniform rows with stored signs and permuted buckets). *Keep the exact product-grid notation and exact loss $\mathcal{L}_{\text{pg, row}}$ from the preceding downstream corollaries. Let the CountSketch signs be stored by*

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

so each stored sign satisfies

$$|\widehat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| \leq u.$$

For each realized exact hash h , fix an exact bucket order

$$k_{i,1}, \dots, k_{i,b_i(h)} \quad \text{of} \quad B_i(h) = \{k : h(k) = i\}.$$

The order is an exact discrete memory-layout input and is not a floating-point quantity.

The exact sparse sketch is

$$Y_{ij} = \sum_{t=1}^{b_i(h)} \omega_{k_i,t} A_{k_i,t,j}.$$

The computed sparse sketch

$$\widehat{Y}^{\rho_{\text{sig}}, \pi} = \text{fl}_{\pi}(S_{h, \widehat{\omega}^{\rho_{\text{sig}}}} A)$$

stores the signs, forms rounded signed products, and accumulates the selected bucket entries in the chosen order π . Its entrywise radius is

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}, \pi} &:= (u + \gamma_{b_i(h)} + u\gamma_{b_i(h)}) \sum_{t=1}^{b_i(h)} |\widehat{\omega}_{k_i,t}^{\rho_{\text{sig}}}| |A_{k_i,t,j}| \\ &\quad + u \sum_{t=1}^{b_i(h)} |A_{k_i,t,j}|. \end{aligned}$$

Let the downstream exact row trace be $\sigma = (\sigma_1, \dots, \sigma_s)$, where each $\sigma_q \in \{1, \dots, r\}$ is drawn from the exact uniform law. Set

$$d = \sqrt{s/r}, \quad \widehat{d}_{\star} = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{r}))),$$

and assume the concrete denominator certificate

$$\widehat{d}_{\star} \neq 0, \quad |\widehat{d}_{\star} - d| \leq d \Delta_{\star}, \quad \Delta_{\star} = \frac{4u + 3u^2 + u^3}{1 - u}.$$

Define

$$B_{qj} = \frac{Y_{\sigma_q j}}{d}, \quad \widetilde{B}_{qj}^{\rho_{\text{sig}}, \pi} = \frac{\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}, \pi}}{d},$$

and

$$\widehat{B}_{qj}^{\rho_{\text{sig}}, \pi} = \text{fl}\left(\frac{\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}, \pi}}{\widehat{d}_{\star}}\right).$$

The computed-denominator row entry radius is

$$E_{qj}^{\text{den}, \rho_{\text{sig}}, \pi} = u |\widehat{B}_{qj}^{\rho_{\text{sig}}, \pi}| + \frac{|\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}, \pi}|}{|\widehat{d}_{\star}|} \Delta_{\star},$$

and the sampled sparse-sketch basis perturbation is

$$F_{qj}^{\rho_{\text{sig}}, \pi} = \frac{E_{\sigma_q j}^{\text{CS}, \rho_{\text{sig}}, \pi}}{d}.$$

Define the deterministic downstream floating-point radius by

$$T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \pi}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}, \pi}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}, \pi}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}, \pi}^{\text{fp}},$$

where

$$T_{\text{den}, \rho_{\text{sig}}, \pi}^{\text{fp}} = \left\| \left(\sum_{q=1}^s E_{qj}^{\text{den}, \rho_{\text{sig}}, \pi} |\widehat{B}_{qk}^{\rho_{\text{sig}}, \pi}| + |\widetilde{B}_{qj}^{\rho_{\text{sig}}, \pi}| E_{qk}^{\text{den}, \rho_{\text{sig}}, \pi} \right)_{j,k=1}^n \right\|_F,$$

$$T_{\text{dot}, \rho_{\text{sig}}, \pi}^{\text{fp}} = \left\| \left(\gamma_s \sum_{q=1}^s |\hat{B}_{qj}^{\rho_{\text{sig}}, \pi}| |\hat{B}_{qk}^{\rho_{\text{sig}}, \pi}| \right)_{j,k=1}^n \right\|_F,$$

and

$$T_{\text{basis}, \rho_{\text{sig}}, \pi}^{\text{fp}} = \left\| \left(\sum_{q=1}^s F_{qj}^{\rho_{\text{sig}}, \pi} |\tilde{B}_{qk}^{\rho_{\text{sig}}, \pi}| + |B_{qj}| F_{qk}^{\rho_{\text{sig}}, \pi} \right)_{j,k=1}^n \right\|_F.$$

With $\eta_{\text{row}} > 0$ and

$$\tau_{\text{CS}} = \eta + L(2\rho + \rho^2),$$

the fully computed sampled Gram

$$\hat{G}_{\text{row}}^{\rho_{\text{sig}}, \pi}(h, \omega, \sigma; A) = \text{fl} \left((\hat{B}^{\rho_{\text{sig}}, \pi})^T \hat{B}^{\rho_{\text{sig}}, \pi} \right)$$

satisfies

$$\begin{aligned} \mathbb{P}_{h, \omega, \sigma} \left[-(\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \pi}^{\text{fp}}) I_n \preceq \hat{G}_{\text{row}}^{\rho_{\text{sig}}, \pi}(h, \omega, \sigma; A) - A^T A \right. \\ \left. \preceq (\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \pi}^{\text{fp}}) I_n \right] \geq 1 - \mathcal{L}_{\text{pg}, \text{row}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg}, \text{row}} \leq \delta$, then the displayed fully computed event has probability at least $1 - \delta$.

The exact objects are A , the product grid z_a , the exact CountSketch hash/sign law, the exact row-sampling law, the exact per-bucket order π , and the exact reference $A^T A$. The computed non-probability objects are $\hat{\omega}^{\rho_{\text{sig}}}$, $\hat{Y}^{\rho_{\text{sig}}, \pi}$, $\hat{d}_\star$, $\hat{B}^{\rho_{\text{sig}}, \pi}$, and $\hat{G}_{\text{row}}^{\rho_{\text{sig}}, \pi}$.

For fixed realized $h, \omega, \sigma, \pi, A$, fixed $d > 0$, and nonzero leading deterministic coefficient,

$$\begin{aligned} T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \pi}^{\text{fp}} = \Theta \left(\gamma_s \left\| |\hat{B}^{\rho_{\text{sig}}, \pi}|^T |\hat{B}^{\rho_{\text{sig}}, \pi}| \right\|_F \right. \\ \left. + \left\| (E^{\text{den}, \rho_{\text{sig}}, \pi})^T |\hat{B}^{\rho_{\text{sig}}, \pi}| + |\tilde{B}^{\rho_{\text{sig}}, \pi}|^T E^{\text{den}, \rho_{\text{sig}}, \pi} \right\|_F \right. \\ \left. + \left\| (F^{\rho_{\text{sig}}, \pi})^T |\tilde{B}^{\rho_{\text{sig}}, \pi}| + |B|^T F^{\rho_{\text{sig}}, \pi} \right\|_F \right). \end{aligned}$$

Substituting the irreducible definitions,

$$E_{qj}^{\text{den}, \rho_{\text{sig}}, \pi} = \Theta \left(u |\hat{B}_{qj}^{\rho_{\text{sig}}, \pi}| + \frac{|\hat{Y}_{\sigma q j}^{\rho_{\text{sig}}, \pi}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u} \right),$$

and

$$F_{qj}^{\rho_{\text{sig}}, \pi} = \Theta \left(\frac{u + \gamma_{b_{\sigma q}(h)}}{d} \sum_{t=1}^{b_{\sigma q}(h)} |A_{k_{\sigma q, t} j}| \right).$$

The deterministic radius tends to zero as

$$u, \gamma_s, \gamma_{b_i(h)}, \frac{4u + 3u^2 + u^3}{1 - u} \rightarrow 0$$

for fixed realized data, while the probability loss remains exactly $\mathcal{L}_{\text{pg}, \text{row}}$.

Corollary 37 (Product-grid CountSketch plus downstream uniform rows with stored signs and tree-reduced buckets). *Let $A \in \mathbb{R}^{m \times n}$ be the exact input matrix, let $r > 0$ be the number of CountSketch rows, and let $s > 0$ be the number of downstream uniformly sampled rows. Let $\mathcal{I}$ be the exact*

finite product-grid index set, let $z_a \in \mathbb{R}^n$ denote the exact grid vector indexed by $a : \{1, \dots, n\} \rightarrow \mathcal{I}$, and let $\eta, L, \rho, \eta_{\text{row}} > 0$. Define the exact product-grid plus downstream row-sampling loss

$$\mathcal{L}_{\text{pg,row}} = \sum_{a:\{1,\dots,n\} \rightarrow \mathcal{I}} \frac{2 \|Az_a\|_2^4}{r\eta^2} + \frac{2 \|A\|_F^4}{rL^2} + \frac{r m^2 \|A\|_F^4}{s\eta_{\text{row}}^2}.$$

The exact CountSketch hash is $h : \{1, \dots, m\} \rightarrow \{1, \dots, r\}$, the exact signs are $\omega_k \in \{-1, 1\}$, and the exact downstream row trace is $\sigma = (\sigma_1, \dots, \sigma_s)$, drawn from the exact iid uniform law on $\{1, \dots, r\}$. These probability laws are exact mathematical inputs.

Let the implemented sign table be formed by one of the concrete routines

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

so that $\hat{\omega}_k^{\rho_{\text{sig}}}$ is respectively $\text{fl}(\omega_k \cdot 1)$, $\text{fl}(\omega_k + 0)$, or $\text{fl}(\omega_k - 0)$, and

$$|\hat{\omega}_k^{\rho_{\text{sig}}} - \omega_k| \leq u.$$

For each realized hash h and bucket i , set

$$B_i(h) = \{k : h(k) = i\}, \quad b_i(h) = |B_i(h)|,$$

and fix an exact enumeration

$$k_{i,1}, \dots, k_{i,b_i(h)}$$

of $B_i(h)$. Let $\mathcal{T}_i(h)$ be an exact binary summation tree with $b_i(h) + 1$ leaves: the first $b_i(h)$ leaves hold the rounded signed products, and the final leaf is an exact zero padding value. Define

$$d_i(h) = \text{depth}(\mathcal{T}_i(h)).$$

Assume the floating-point model has valid $\gamma_{d_i(h)}$ for every realized bucket tree and valid γ_s for the downstream length- s sampled-Gram dot products.

The exact sparse sketch is

$$Y_{ij} = \sum_{t=1}^{b_i(h)} \omega_{k_{i,t}} A_{k_{i,t}j}.$$

The computed sparse sketch

$$\hat{Y}^{\rho_{\text{sig}}, \mathcal{T}}$$

is obtained by storing the signs, forming rounded products $\text{fl}(\hat{\omega}_{k_{i,t}}^{\rho_{\text{sig}}} A_{k_{i,t}j})$, and reducing each bucket through $\mathcal{T}_i(h)$. Its entrywise radius is

$$\begin{aligned} E_{ij}^{\text{CS}, \rho_{\text{sig}}, \mathcal{T}} &:= (u + \gamma_{d_i(h)} + u\gamma_{d_i(h)}) \sum_{t=1}^{b_i(h)} |\hat{\omega}_{k_{i,t}}^{\rho_{\text{sig}}} A_{k_{i,t}j}| \\ &\quad + u \sum_{t=1}^{b_i(h)} |A_{k_{i,t}j}|. \end{aligned}$$

Set the exact row-scale denominator and the concrete computed denominator to

$$d = \sqrt{s/r}, \quad \hat{d}_\star = \text{fl}(\text{fl}(\sqrt{s}) \text{fl}(1/\text{fl}(\sqrt{r}))).$$

The concrete denominator certificate is

$$\widehat{d}_\star \neq 0, \quad |\widehat{d}_\star - d| \leq d \Delta_\star, \quad \Delta_\star = \frac{4u + 3u^2 + u^3}{1 - u}.$$

For $q = 1, \dots, s$ and $j = 1, \dots, n$, define

$$B_{qj} = \frac{Y_{\sigma_{qj}}}{d}, \quad \widetilde{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}} = \frac{\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}, \mathcal{T}}}{d},$$

and

$$\widehat{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}} = \text{fl} \left(\frac{\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}, \mathcal{T}}}{\widehat{d}_\star} \right).$$

The computed-denominator row entry radius is

$$E_{qj}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} = u |\widehat{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}}| + \frac{|\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}, \mathcal{T}}|}{|\widehat{d}_\star|} \Delta_\star,$$

and the sampled sparse-sketch basis perturbation is

$$F_{qj}^{\rho_{\text{sig}}, \mathcal{T}} = \frac{E_{\sigma_{qj}}^{\text{CS}, \rho_{\text{sig}}, \mathcal{T}}}{d}.$$

Define the deterministic downstream floating-point radius

$$T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}},$$

where

$$T_{\text{den}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = \left\| \left(\sum_{q=1}^s E_{qj}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} |\widehat{B}_{qk}^{\rho_{\text{sig}}, \mathcal{T}}| + |\widetilde{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}}| E_{qk}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} \right)_{j,k=1}^n \right\|_F,$$

$$T_{\text{dot}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = \left\| \left(\gamma_s \sum_{q=1}^s |\widehat{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}}| |\widehat{B}_{qk}^{\rho_{\text{sig}}, \mathcal{T}}| \right)_{j,k=1}^n \right\|_F,$$

and

$$T_{\text{basis}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}} = \left\| \left(\sum_{q=1}^s F_{qj}^{\rho_{\text{sig}}, \mathcal{T}} |\widetilde{B}_{qk}^{\rho_{\text{sig}}, \mathcal{T}}| + |B_{qj}| F_{qk}^{\rho_{\text{sig}}, \mathcal{T}} \right)_{j,k=1}^n \right\|_F.$$

With

$$\tau_{\text{CS}} = \eta + L(2\rho + \rho^2),$$

the fully computed sampled Gram

$$\widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \mathcal{T}}(h, \omega, \sigma; A) = \text{fl} \left((\widehat{B}^{\rho_{\text{sig}}, \mathcal{T}})^T \widehat{B}^{\rho_{\text{sig}}, \mathcal{T}} \right)$$

satisfies

$$\begin{aligned} \mathbb{P}_{h, \omega, \sigma} \left[-(\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \mathcal{T}}(h, \omega, \sigma; A) - A^T A \right. \\ \left. \preceq (\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \mathcal{T}}^{\text{fp}}) I_n \right] \geq 1 - \mathcal{L}_{\text{pg}, \text{row}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg,row}} \leq \delta$, then the displayed fully computed event has probability at least $1 - \delta$.

The exact objects are A , the product grid z_a , the exact CountSketch hash/sign law, the exact downstream row-sampling law, the exact tree shapes $\mathcal{T}_i(h)$, and the reference $A^T A$. The tree shapes are exact discrete implementation choices and not floating-point real computations. The computed non-probability objects are $\hat{\omega}^{\rho_{\text{sig}}}$, $\hat{Y}^{\rho_{\text{sig}}, \mathcal{T}}$, $\hat{d}_\star$, $\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}$, and $\hat{G}_{\text{row}}^{\rho_{\text{sig}}, \mathcal{T}}$. No extra perturbation event or extra existence hypothesis is used.

For fixed realized $h, \omega, \sigma, \mathcal{T}, A$, fixed $d > 0$, and nonzero leading deterministic coefficient,

$$\begin{aligned} T_{\text{CS,row},\rho_{\text{sig}},\mathcal{T}}^{\text{fp}} = & \Theta \left(\gamma_s \left\| |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}|^T |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}| \right\|_F \right. \\ & + \left\| (E^{\text{den}, \rho_{\text{sig}}, \mathcal{T}})^T |\hat{B}^{\rho_{\text{sig}}, \mathcal{T}}| + |\tilde{B}^{\rho_{\text{sig}}, \mathcal{T}}|^T E^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} \right\|_F \\ & \left. + \left\| (F^{\rho_{\text{sig}}, \mathcal{T}})^T |\tilde{B}^{\rho_{\text{sig}}, \mathcal{T}}| + |B|^T F^{\rho_{\text{sig}}, \mathcal{T}} \right\|_F \right). \end{aligned}$$

Substituting the irreducible definitions,

$$E_{qj}^{\text{den}, \rho_{\text{sig}}, \mathcal{T}} = \Theta \left(u |\hat{B}_{qj}^{\rho_{\text{sig}}, \mathcal{T}}| + \frac{|\hat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}, \mathcal{T}}|}{|\hat{d}_\star|} \frac{4u + 3u^2 + u^3}{1 - u} \right),$$

and

$$F_{qj}^{\rho_{\text{sig}}, \mathcal{T}} = \Theta \left(\frac{u + \gamma_{d_{\sigma_q}(h)}}{d} \sum_{t=1}^{b_{\sigma_q}(h)} |A_{k_{\sigma_q, t} j}| \right).$$

If the selected bucket trees are balanced, so that

$$d_i(h) = \Theta(\log(b_i(h) + 1)),$$

then in the usual small- u regime

$$F_{qj}^{\rho_{\text{sig}}, \mathcal{T}} = \Theta \left(\frac{u(1 + \log(b_{\sigma_q}(h) + 1))}{d} \sum_{t=1}^{b_{\sigma_q}(h)} |A_{k_{\sigma_q, t} j}| \right).$$

The deterministic radius tends to zero as

$$u, \gamma_s, \gamma_{d_i}(h), \Delta_\star \rightarrow 0$$

for fixed realized data, while the probability loss remains exactly $\mathcal{L}_{\text{pg,row}}$.

Corollary 38 (Stored-sign CountSketch with alternate computed denominators). *Keep the hypotheses, exact product-grid loss $\mathcal{L}_{\text{pg,row}}$, stored-sign mode*

$$\rho_{\text{sig}} \in \{\text{mul1}, \text{add0}, \text{sub0}\},$$

and exact denominator $d = \sqrt{s/r}$ from the preceding stored-sign corollary. Let the denominator route be

$$\lambda \in \{\text{sqrtExact}, \text{divSqrt}, \text{invMulSqrt}, \text{sqrtDivSqrt}\}.$$

The corresponding computed denominators are

$$\begin{aligned} \lambda = \text{sqrtExact} : \quad & \hat{d}_\lambda = \text{fl}_{\text{sqrt}}(s/r), \quad \text{where the input ratio } s/r \text{ is supplied exactly,} \\ \lambda = \text{divSqrt} : \quad & \hat{d}_\lambda = \text{fl}_{\text{sqrt}}(\text{fl}_{\text{div}}(s, r)), \\ \lambda = \text{invMulSqrt} : \quad & \hat{d}_\lambda = \text{fl}_{\text{sqrt}}(\text{fl}_{\text{mul}}(s, \text{fl}_{\text{div}}(1, r))), \\ \lambda = \text{sqrtDivSqrt} : \quad & \hat{d}_\lambda = \text{fl}_{\text{div}}(\text{fl}_{\text{sqrt}}(s), \text{fl}_{\text{sqrt}}(r)). \end{aligned}$$

For each route the concrete denominator certificate proves

$$\widehat{d}_\lambda \neq 0, \quad |\widehat{d}_\lambda - d| \leq d \Delta_\lambda,$$

where

$$\begin{aligned} \Delta_{\text{sqrtExact}} &= u, & \Delta_{\text{divSqrt}} &= \sqrt{1+u}u + u, \\ \Delta_{\text{invMulSqrt}} &= \sqrt{(1+u)(1+u)}u + 2u + u^2, & \Delta_{\text{sqrtDivSqrt}} &= \frac{3u + u^2}{1-u}. \end{aligned}$$

Define the route-dependent scaled rows

$$\widehat{B}_{qj}^{\rho_{\text{sig}}, \lambda} = \text{fl} \left(\frac{\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}}}{\widehat{d}_\lambda} \right), \quad \widetilde{B}_{qj}^{\rho_{\text{sig}}} = \frac{\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}}}{d}.$$

The exact reference $B_{qj} = Y_{\sigma_{qj}}/d$ is unchanged. The route-dependent computed-denominator row entry radius is

$$E_{qj}^{\text{den}, \rho_{\text{sig}}, \lambda} = u |\widehat{B}_{qj}^{\rho_{\text{sig}}, \lambda}| + \frac{|\widehat{Y}_{\sigma_{qj}}^{\rho_{\text{sig}}}|}{|\widehat{d}_\lambda|} \Delta_\lambda.$$

The stored-sign sparse-apply sampled radius is still

$$F_{qj}^{\rho_{\text{sig}}} = \frac{E_{\sigma_{qj}}^{\text{CS}, \rho_{\text{sig}}}}{d}.$$

Set

$$T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \lambda}^{\text{fp}} = T_{\text{den}, \rho_{\text{sig}}, \lambda}^{\text{fp}} + T_{\text{dot}, \rho_{\text{sig}}, \lambda}^{\text{fp}} + T_{\text{basis}, \rho_{\text{sig}}, \lambda}^{\text{fp}},$$

with

$$\begin{aligned} T_{\text{den}, \rho_{\text{sig}}, \lambda}^{\text{fp}} &= \left\| \left(\sum_{q=1}^s E_{qj}^{\text{den}, \rho_{\text{sig}}, \lambda} |\widehat{B}_{qk}^{\rho_{\text{sig}}, \lambda}| + |\widetilde{B}_{qj}^{\rho_{\text{sig}}}| E_{qk}^{\text{den}, \rho_{\text{sig}}, \lambda} \right)_{j,k=1}^n \right\|_F, \\ T_{\text{dot}, \rho_{\text{sig}}, \lambda}^{\text{fp}} &= \left\| \left(\gamma_s \sum_{q=1}^s |\widehat{B}_{qj}^{\rho_{\text{sig}}, \lambda}| |\widehat{B}_{qk}^{\rho_{\text{sig}}, \lambda}| \right)_{j,k=1}^n \right\|_F, \end{aligned}$$

and

$$T_{\text{basis}, \rho_{\text{sig}}, \lambda}^{\text{fp}} = \left\| \left(\sum_{q=1}^s F_{qj}^{\rho_{\text{sig}}} |\widetilde{B}_{qk}^{\rho_{\text{sig}}}| + |B_{qj}| F_{qk}^{\rho_{\text{sig}}} \right)_{j,k=1}^n \right\|_F.$$

Then the fully computed stored-sign downstream sampled Gram

$$\widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \lambda}(h, \omega, \sigma; A) = \text{fl} \left((\widehat{B}^{\rho_{\text{sig}}, \lambda})^T \widehat{B}^{\rho_{\text{sig}}, \lambda} \right)$$

satisfies

$$\begin{aligned} \mathbb{P}_{h, \omega, \sigma} \left[-(\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \lambda}^{\text{fp}}) I_n \preceq \widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \lambda}(h, \omega, \sigma; A) - A^T A \right. \\ \left. \preceq (\tau_{\text{CS}} + \eta_{\text{row}} + T_{\text{CS}, \text{row}, \rho_{\text{sig}}, \lambda}^{\text{fp}}) I_n \right] \geq 1 - \mathcal{L}_{\text{pg}, \text{row}}. \end{aligned}$$

Equivalently, if $\mathcal{L}_{\text{pg}, \text{row}} \leq \delta$, then the displayed fully computed event has probability at least $1 - \delta$.

The exact objects are A , the product grid z_a , the exact CountSketch hash/sign law, the exact uniform row-sampling law, and the exact reference matrix $A^T A$. The computed non-probability objects are $\widehat{\omega}^{\rho_{\text{sig}}}$, $\widehat{Y}^{\rho_{\text{sig}}}$, $\widehat{d}_\lambda$, $\widehat{B}^{\rho_{\text{sig}}, \lambda}$, and $\widehat{G}_{\text{row}}^{\rho_{\text{sig}}, \lambda}$.

For fixed realized h, ω, σ, A , fixed $d > 0$, and nonzero leading deterministic coefficient,

$$\begin{aligned} T_{\text{CS, row}, \rho_{\text{sig}}, \lambda}^{\text{fp}} = & \Theta \left(\gamma_s \left\| |\widehat{B}^{\rho_{\text{sig}}, \lambda}|^T |\widehat{B}^{\rho_{\text{sig}}, \lambda}| \right\|_F \right. \\ & + \left\| (E^{\text{den}, \rho_{\text{sig}}, \lambda})^T |\widehat{B}^{\rho_{\text{sig}}, \lambda}| + |\widetilde{B}^{\rho_{\text{sig}}}|^T E^{\text{den}, \rho_{\text{sig}}, \lambda} \right\|_F \\ & \left. + \left\| (F^{\rho_{\text{sig}}})^T |\widetilde{B}^{\rho_{\text{sig}}}| + |B|^T F^{\rho_{\text{sig}}} \right\|_F \right), \end{aligned}$$

and, after substituting the irreducible definitions,

$$\begin{aligned} E_{qj}^{\text{den}, \rho_{\text{sig}}, \lambda} &= \Theta \left(u |\widehat{B}_{qj}^{\rho_{\text{sig}}, \lambda}| + \frac{|\widehat{Y}_{\sigma_q j}^{\rho_{\text{sig}}}|}{|\widehat{d}_\lambda|} \Delta_\lambda \right), \\ F_{qj}^{\rho_{\text{sig}}} &= \Theta \left(\frac{u + \gamma_{b_{\sigma_q}(h)}}{d} \sum_{k: h(k) = \sigma_q} |A_{kj}| \right). \end{aligned}$$

Thus the deterministic floating-point radius is non-vacuous in the small roundoff regime and tends to zero as

$$u, \gamma_s, \gamma_{b_i(h)}, \Delta_\lambda \rightarrow 0.$$

The probability-loss order is unchanged from the exact product-grid CountSketch plus exact down-stream row-sampling loss; denominator computation affects only the deterministic radius.

References

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